

Probability and Statistics Assignments

Prob Stats 1: Assignment Solutions

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Chapter 1

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Assignment 1

Prob Stats 1: Assignment Solutions

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1. From a group of 5 women and 7 men, a committee of 7 is to be formed randomly.

(a) The probability that the committee consists of 3 women and 4 men is -

$$P = \frac{\binom{5}{3} \cdot \binom{7}{4}}{\binom{12}{7}} = \frac{10 \cdot 35}{792} = 0.441 \dots$$

(b) Let the probability that the committee contains Mr. X be $P(A)$. To form a committee containing Mr. X we choose the remaining 6 members from the other 11 persons, hence

$$P(A) = \frac{\binom{11}{6}}{\binom{12}{7}} = \frac{462}{792} = \frac{7}{12} = 0.5833 \dots$$

(c) The probability that Mr. X is not included is

$$P(A^C) = 1 - P(A) = 1 - \frac{7}{12} = \frac{5}{12} = 0.4166 \dots$$

(Equivalently $P(A^C) = \frac{\binom{11}{7}}{\binom{12}{7}} = \frac{330}{792} = \frac{5}{12}$.)

(d) Suppose two particular men refuse to serve together. Total number of committees is $\binom{12}{7} = 792$. Number of committees that include both refusing men is: choose the remaining 5 members from the other 10 people,

$$\binom{10}{5} = 252.$$

So the number of workable committees (those not containing both) is $792 - 252 = 540$, and the required probability is

$$\frac{540}{792} = \frac{15}{22} = 0.681818 \dots$$

2. If 5 people are present in a room, assuming 365 possible birthdays and independence,

(a) Probability no two have the same birthday:

$$P(\text{all distinct}) = \frac{365 \cdot 364 \cdot 363 \cdot 362 \cdot 361}{365^5}$$

(b) Probability exactly two persons share a birthday (and the other three all have distinct birthdays different from that one):

$$P(\text{exactly one pair}) = \binom{5}{2} \cdot 365 \cdot \frac{364 \cdot 363 \cdot 362}{365^3}$$

Here $\binom{5}{2}$ chooses which two form the pair, 365 chooses the common birthday for the pair, and $P(364, 3) = 364 \cdot 363 \cdot 362$ assigns distinct birthdays to the remaining three from the remaining days.

3. Sanjit, Prashant and Bharath have probabilities 0.8, 0.7, 0.6 respectively to solve a problem, independently.

(a) Only Sanjit solves it:

$$P(\text{only S}) = 0.8 \cdot (1 - 0.7) \cdot (1 - 0.6) = 0.8 \cdot 0.3 \cdot 0.4 = 0.096$$

(b) Only Prashant solves it:

$$P(\text{only P}) = (1 - 0.8) \cdot 0.7 \cdot (1 - 0.6) = 0.2 \cdot 0.7 \cdot 0.4 = 0.056$$

(c) Only Bharath solves it:

$$P(\text{only B}) = (1 - 0.8) \cdot (1 - 0.7) \cdot 0.6 = 0.2 \cdot 0.3 \cdot 0.6 = 0.036$$

(d) If it is known that the problem is solved (i.e. at least one solved it), first compute

$$P(\text{solved}) = 1 - P(\text{none}) = 1 - (1 - 0.8)(1 - 0.7)(1 - 0.6) = 1 - 0.024 = 0.976$$

Then conditional probabilities become

$$P(\text{only S} \mid \text{solved}) = \frac{0.096}{0.976} \approx 0.09836, \quad P(\text{only P} \mid \text{solved}) = \frac{0.056}{0.976} \approx 0.05738,$$

$$P(\text{only B} \mid \text{solved}) = \frac{0.036}{0.976} \approx 0.03689.$$

4. In a modelling agency: $P(\text{under } 22) = \frac{2}{3}$, $P(\text{male}) = \frac{3}{5}$, and $P(\text{female or } 22^+) = \frac{5}{8}$. Find $P(\text{female and under } 22)$.

Let U = under 22, O = 22 or older, F = female. Then $P(U) = \frac{2}{3}$, $P(O) = 1 - P(U) = \frac{1}{3}$, and $P(F) = 1 - P(\text{male}) = 1 - \frac{3}{5} = \frac{2}{5}$. Using

$$P(F \cap O) = P(F) + P(O) - P(F \cup O) = \frac{2}{5} + \frac{1}{3} - \frac{5}{8}$$

compute with common denominator 120:

$$P(F \cap O) = \frac{48 + 40 - 75}{120} = \frac{13}{120}$$

Therefore -

$$P(F \cap U) = P(F) - P(F \cap O) = \frac{2}{5} - \frac{13}{120} = \frac{48 - 13}{120} = \frac{35}{120} = \frac{7}{24} \approx 0.29167$$

5. If A and B are independent, show:

(a) A and B^c are independent:

$$P(A \cap B^c) = P(A) - P(A \cap B) = P(A) - P(A)P(B) = P(A)(1 - P(B)) = P(A)P(B^c).$$

(b) A^c and B are independent: analogous to (a).

(c) A^c and B^c are independent: likewise,

$$P(A^c \cap B^c) = 1 - P(A \cup B) = 1 - (P(A) + P(B) - P(A)P(B)) = (1 - P(A))(1 - P(B)) = P(A^c)P(B^c).$$

6. Show that pairwise independence may not imply mutual independence.

Example: Let the sample space be $\{1, 2, 3, 4\}$ with each outcome probability $\frac{1}{4}$. Define

$$A = \{1, 2\}, \quad B = \{1, 3\}, \quad C = \{1, 4\}.$$

Then $P(A) = P(B) = P(C) = \frac{1}{2}$ and for each pair

$$P(A \cap B) = P(A \cap C) = P(B \cap C) = P(\{1\}) = \frac{1}{4} = P(A)P(B).$$

Thus the three events are pairwise independent. But

$$P(A \cap B \cap C) = P(\{1\}) = \frac{1}{4} \neq \frac{1}{8} = P(A)P(B)P(C),$$

so they are not mutually independent.

7. Two persons A and B roll an unbiased die alternately. The person who gets a 6 first wins.

(a) If A rolls first, the probability A wins is

$$P(A \text{ wins}) = \sum_{k=0}^{\infty} (\text{no 6 in first } 2k \text{ rolls}) \cdot P(A \text{ gets 6 next}) = \sum_{k=0}^{\infty} \left(\frac{5}{6}\right)^{2k} \cdot \frac{1}{6}$$

This is a geometric series with sum

$$P(A \text{ wins}) = \frac{1/6}{1 - (5/6)^2} = \frac{1/6}{1 - 25/36} = \frac{1/6}{11/36} = \frac{6}{11}$$

(b) If B rolls first, by symmetry $P(B \text{ wins}) = \frac{6}{11}$ and hence $P(A \text{ wins}) = \frac{5}{11}$.

8. Justify:

(a) A probability mass function (pmf) $p(x)$ for a discrete random variable satisfies $\sum_x p(x) = 1$ and $p(x) \geq 0$, so each $p(x) \leq 1$ (a term in a sum that equals 1 cannot exceed 1). A probability density function (pdf) $f(x)$ for a continuous variable integrates to 1 but can exceed 1 on sets of small measure; e.g. $f(x) = 2$ for $x \in [0, \frac{1}{2}]$ has $\int_0^{1/2} 2 dx = 1$ but $f(x) = 2 > 1$ on that interval.

9. Show that

$$(a) P\left(\bigcap_{i=1}^n A_i\right) \geq 1 - \sum_{i=1}^n P(A_i^c).$$

Proof. We have -

$$P\left(\bigcap_{i=1}^n A_i\right) = 1 - P\left(\bigcup_{i=1}^n A_i^c\right) \geq 1 - \sum_{i=1}^n P(A_i^c) \quad \left[P\left(\bigcup_i E_i\right) \leq \sum_i P(E_i) \right]$$

□

$$(b) P\left(\bigcap_{i=1}^n A_i\right) \geq \sum_{i=1}^n P(A_i) - (n-1).$$

Proof. Using the previous inequality with A_i replaced by A_i^c ,

$$P\left(\bigcap_{i=1}^n A_i\right) = 1 - P\left(\bigcup_{i=1}^n A_i^c\right) \geq 1 - \sum_{i=1}^n P(A_i^c)$$

But $\sum_{i=1}^n P(A_i^c) = n - \sum_{i=1}^n P(A_i)$, so

$$P\left(\bigcap_{i=1}^n A_i\right) \geq 1 - (n - \sum_{i=1}^n P(A_i)) = \sum_{i=1}^n P(A_i) - (n-1)$$

□

10. Multiple choice test: let K be event "student knew the answer" with $P(K) = p$, and a guess (if not knowing) is correct with probability $1/m$. Find $P(K | \text{correct})$.

We have

$$P(\text{correct}) = P(K) \cdot 1 + P(K^c) \cdot \frac{1}{m} = p + \frac{1-p}{m}$$

Since $P(K \cap \text{correct}) = P(K) = p$, by Bayes' rule -

$$P(K | \text{correct}) = \frac{p}{p + \frac{1-p}{m}} = \frac{p}{p + (1-p)/m}$$

Assignment 2
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1. Let A be the event that the disease is present, B be the event that the test is positive. We are given -

$$P(A) = 0.005, \quad P(B | A) = 0.95, \quad P(B | A^C) = 0.01$$

We need to find $P(A | B)$. By Bayes' theorem,

$$P(A | B) = \frac{P(B | A)P(A)}{P(B | A)P(A) + P(B | A^C)P(A^C)} = \frac{0.95 \cdot 0.005}{0.95 \cdot 0.005 + 0.01 \cdot 0.995}$$

Calculating the denominator,

$$0.95 \cdot 0.005 + 0.01 \cdot 0.995 = 0.00475 + 0.00995 = 0.0147$$

Thus,

$$P(A | B) = \frac{0.00475}{0.0147} \approx 0.3231$$

2. We have -

$$P(A^C) = 0.3, \quad P(B) = 0.4, \quad P(A \cap B^C) = 0.5$$

We need to find $P(B | A \cup B^C)$. First, we compute $P(A \cup B^C)$:

$$P(A \cup B^C) = 1 - P(A^C \cap B) = 1 - [P(B) - P(A \cap B)]$$

We know $P(A \cap B^C) = P(A) - P(A \cap B)$, so

$$P(A \cap B) = P(A) - P(A \cap B^C) = 0.7 - 0.5 = 0.2$$

Thus,

$$P(A^C \cap B) = P(B) - P(A \cap B) = 0.4 - 0.2 = 0.2$$

Therefore,

$$P(A \cup B^C) = 1 - 0.2 = 0.8$$

Now, we can find $P(B | A \cup B^C)$:

$$P(B | A \cup B^C) = \frac{P(B \cap (A \cup B^C))}{P(A \cup B^C)} = \frac{P(A \cap B)}{P(A \cup B^C)} = \frac{0.2}{0.8} = 0.25$$

3. We have -

$$P(A) = 0.35, \quad P(B) = 0.15, \quad P(C) = 0.3, \quad P(D) = 0.2$$

We have that B has withdrawn. The probability that the other companies are awarded the contract are now - $P(A | B^C)$, $P(C | B^C)$, and $P(D | B^C)$. Since the events are mutually exclusive and exhaustive, we have

$$P(B^C) = 1 - P(B) = 0.85$$

Thus,

$$P(A | B^C) = \frac{P(A)}{P(B^C)} = \frac{0.35}{0.85} \approx 0.4118$$

$$P(C | B^C) = \frac{P(C)}{P(B^C)} = \frac{0.3}{0.85} \approx 0.3529$$

$$P(D | B^C) = \frac{P(D)}{P(B^C)} = \frac{0.2}{0.85} \approx 0.2353$$

4. Let the events be defined as follows: A : Red unit meets its objectives, B : Blue unit meets its objectives. We have -

$$P(A) = 0.6, \quad P(B) = 0.7, \quad P(A \cap B^C) = 0.18$$

We need to find $P(A \cap B)$. We also need to find $P(A \cap B^C), P(B \cap A^C)$ to find the probability that only one unit meets its objectives, that is $P(A \cap B^C) + P(B \cap A^C)$ We have -

$$P(A) = P(A \cap B) + P(A \cap B^C)$$

Thus,

$$P(A \cap B) = P(A) - P(A \cap B^C) = 0.6 - 0.18 = 0.42$$

Similarly,

$$P(B) = P(B \cap A) + P(B \cap A^C)$$

Thus,

$$P(B \cap A^C) = P(B) - P(A \cap B) = 0.7 - 0.42 = 0.28$$

Therefore, the probability that only one unit meets its objectives is -

$$P(A \cap B^C) + P(B \cap A^C) = 0.18 + 0.28 = 0.46$$

5. Let the events be defined as follows: A : People read newspaper I, B : People read newspaper II, C : People read newspaper III. We have -

$$P(A) = 0.1, \quad P(B) = 0.3, \quad P(C) = 0.05$$

Further -

$$P(A \cap B) = 0.08, \quad P(A \cap C) = 0.02, \quad P(B \cap C) = 0.04, \quad P(A \cap B \cap C) = 0.01$$

- (a) People who read only one newspaper:

$$P(\text{only } A) = P(A) - P(A \cap B) - P(A \cap C) + P(A \cap B \cap C)$$

$$= 0.1 - 0.08 - 0.02 + 0.01 = 0.01$$

$$P(\text{only } B) = P(B) - P(A \cap B) - P(B \cap C) + P(A \cap B \cap C)$$

$$= 0.3 - 0.08 - 0.04 + 0.01 = 0.19$$

$$P(\text{only } C) = P(C) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$

$$= 0.05 - 0.02 - 0.04 + 0.01 = 0.0$$

Thus, the total probability of people who read only one newspaper is -

$$P(\text{only } A) + P(\text{only } B) + P(\text{only } C) = 0.01 + 0.19 + 0.0 = 0.2$$

- (b) People who read at least two newspapers:

$$P(\text{at least two}) = P(A \cap B) + P(A \cap C) + P(B \cap C) - 2P(A \cap B \cap C)$$

$$= 0.08 + 0.02 + 0.04 - 2 \cdot 0.01 = 0.12$$

Thus, the total probability of people who read at least two newspapers is -

$$P(\text{at least two}) = 0.12$$

- (c) If I and II are morning papers and III is an evening paper, the probability that a randomly chosen person reads at least one morning paper and one evening paper is -

$$\begin{aligned} P((A \cup B) \cap C) &= P(A \cap C) + P(B \cap C) - P(A \cap B \cap C) \\ &= 0.02 + 0.04 - 0.01 = 0.05 \end{aligned}$$

Thus,

$$P(\text{at least one morning and one evening}) = 0.05$$

- (d) People who do not read any newspaper:

$$P(\text{none}) = 1 - P(A \cup B \cup C)$$

Using inclusion-exclusion,

$$\begin{aligned} P(A \cup B \cup C) &= P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C) \\ &= 0.1 + 0.3 + 0.05 - 0.08 - 0.02 - 0.04 + 0.01 = 0.32 \end{aligned}$$

Thus,

$$P(\text{none}) = 1 - 0.32 = 0.68$$

- (e) People who read exactly one morning paper and the evening paper:

$$\begin{aligned} P(\text{exactly one morning and evening}) &= P(\text{only } A \cap C) + P(\text{only } B \cap C) \\ &= (P(A \cap C) - P(A \cap B \cap C)) + (P(B \cap C) - P(A \cap B \cap C)) \\ &= (0.02 - 0.01) + (0.04 - 0.01) = 0.04 \end{aligned}$$

Thus,

$$P(\text{exactly one morning and evening}) = 0.04$$

6. From a truly random digit generator, digits 0-9 are generated with equal probability.

- (a) The total number of possible random two digit combinations is total number of combinations without the first digit being zero, i.e. $9 \times 10 = 90$.
- (b) The probability that a number begins with 2 is $\frac{1}{9}$ since there are 9 possible first digits (1-9).
- (c) For each starting digit (1-9), we have 1 number that ends with 9. Therefore we have 9 such numbers. Thus the probability that a number ends with 9 is $\frac{9}{90} = \frac{1}{10}$.
- (d) The only number that begins with 2 and ends with 9 is 29. Therefore the probability is $\frac{1}{90}$.
- (e) The probability that a randomly formed number ends with 9 given that it begins with 2 is $P(\text{ends with 9} \mid \text{begins with 2}) = \frac{1}{10}$ since for a number beginning with 2, there are 10 possible endings (0-9).

7. Let the events be defined as follows: A : Failure due to transformer damage, B : Failure due to transmission line damage, C : Both problems cause failure. We have -

$$P(A) = 0.05, \quad P(B) = 0.8, \quad P(C) = 0.01$$

- (a) We need to find $P(B \mid A)$. We have -

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)} = \frac{P(C)}{P(A)} = \frac{0.01}{0.05} = 0.2$$

- (b) We need to find $P(A \mid B)$. We have -

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{P(C)}{P(B)} = \frac{0.01}{0.8} = 0.0125$$

- (c) We need to find $P(A \cap B^C)$. We have -

$$P(A \cap B^C) = P(A) - P(A \cap B) = P(A) - P(C) = 0.05 - 0.01 = 0.04$$

(d) We need to find $P(A \cup B)$. Using inclusion-exclusion, we have -

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = P(A) + P(B) - P(C) = 0.05 + 0.8 - 0.01 = 0.84$$

8. Let A be the event that a gene for determining the Rh factor is negative. Let B be the event that individual has negative Rh factor. We have -

$$P(A) = 0.39$$

We know that an individual has negative Rh factor if both genes are negative. Thus,

$$P(B) = P(A \cap A) = P(A)^2 = (0.39)^2 = 0.1521$$

Therefore, the probability that an individual has negative Rh factor is -

$$\boxed{P(B) = 0.1521}$$

9. Let the events be defined as follows: A : Soil has high copper content, B : Mint indicating copper is present. We have -

$$P(A) = 0.3, \quad P(B) = 0.23, \quad P(B | A) = 0.7$$

(a) We need to find $P(A \cap B)$. We have -

$$P(A \cap B) = P(B | A) \cdot P(A) = 0.7 \cdot 0.3 = 0.21$$

(b) We need to find $P(B | A)$. Using Bayes' theorem, we have -

$$P(A | B) = \frac{P(B | A) \cdot P(A)}{P(B)} = \frac{0.7 \cdot 0.3}{0.23} \approx 0.9130$$

10. Let the events be defined as follows: A : Person is accident prone, B : Person gets into an accident. We have -

$$P(A \cap B) = 0.4, \quad P(A^C \cap B) = 0.2, \quad P(A) = 0.3$$

(a) We need to find $P(B)$. We have -

$$P(B) = P(A \cap B) + P(A^C \cap B) = 0.4 + 0.2 = 0.6$$

(b) We need to find $P(A | B)$. Using Bayes' theorem, we have -

$$P(A | B) = \frac{P(A \cap B)}{P(B)} = \frac{0.4}{0.6} = \frac{2}{3} \approx 0.6667$$

Assignment 3
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1. Two fair dice are rolled. The events are defined as follows: E_1 : Sum of the numbers on the two dice is 6, E_2 : Sum of the numbers on the two dice is 7, F : The number on the first die is 4, G : The number on the second die is 3.

(a) E_1 and F are not independent:

$$P(E_1) = \frac{5}{36}, \quad P(F) = \frac{1}{6}, \quad P(E_1 \cap F) = \frac{1}{36}$$
$$P(E_1) \cdot P(F) = \frac{5}{36} \cdot \frac{1}{6} = \frac{5}{216} \neq P(E_1 \cap F)$$

Thus, E_1 and F are not independent.

(b) E_2 and F are independent:

$$P(E_2) = \frac{6}{36} = \frac{1}{6}, \quad P(F) = \frac{1}{6}, \quad P(E_2 \cap F) = \frac{1}{36}$$
$$P(E_2) \cdot P(F) = \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36} = P(E_2 \cap F)$$

Thus, E_2 and F are independent.

(c) E_1 and G are not independent:

$$P(E_1) = \frac{5}{36}, \quad P(G) = \frac{1}{6}, \quad P(E_1 \cap G) = \frac{1}{36}$$
$$P(E_1) \cdot P(G) = \frac{5}{36} \cdot \frac{1}{6} = \frac{5}{216} \neq P(E_1 \cap G)$$

Thus, E_1 and G are not independent.

(d) E_2 and G are independent:

$$P(E_2) = \frac{6}{36} = \frac{1}{6}, \quad P(G) = \frac{1}{6}, \quad P(E_2 \cap G) = \frac{1}{36}$$
$$P(E_2) \cdot P(G) = \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36} = P(E_2 \cap G)$$

Thus, E_2 and G are independent.

(e) E_2 and $F \cap G$ are not independent:

$$P(E_2) = \frac{6}{36} = \frac{1}{6}, \quad P(F \cap G) = \frac{1}{36}, \quad P(E_2 \cap (F \cap G)) = \frac{1}{36}$$
$$P(E_2) \cdot P(F \cap G) = \frac{1}{6} \cdot \frac{1}{36} = \frac{1}{216} \neq P(E_2 \cap (F \cap G))$$

Thus, E_2 and $F \cap G$ are not independent.

2. We have A and B are partition of a sample space. This means that $A \cap B = \emptyset$ and $A \cup B = S$. We also have $A = B^C$ since there are only 2 elements in the partition. Now -

- (a) We need to calculate $P(A | B) + P(A | B^C)$. We have -

$$P(A | B) + P(A | B^C) = \frac{P(A \cap B)}{P(B)} + P(A | A) = 0 + 1 = 1$$

- (b) We need to calculate $P(B | A) + P(A^C | B)$. We have -

$$P(B | A) + P(A^C | B) = \frac{P(B \cap A)}{P(A)} + P(A^C | A^C) = 0 + 1 = 1$$

We can say that the conditional probabilities commute in this case because A and B are complements of each other.

- (c) If A and B were any two events (not necessarily a partition), the above results would not hold. The events A and B being a partition is crucial to the results in (a) and (b) since it ensures that $A \cap B = \emptyset$ and $A \cup B = S$. This property allows us to simplify the conditional probabilities in (a) and (b) to yield the results of 1. If A and B were not a partition, we could not make these simplifications and the results would not hold in general.

3. Let the events be defined as follows: A : Flash flood warning issued, B : Dam Failure during flood. We have -

$$P(A) = 0.5, \quad P(B | A) = 0.17$$

We need to find $P(A \cap B)$. We have -

$$P(A \cap B) = P(B | A) \cdot P(A) = 0.17 \cdot 0.5 = 0.085$$

4. Let the events be defined as follows: A : Dimension A meets specifications, B : Dimension B meets specifications. We have -

$$P(B | A) = 0.98, \quad P(A) = 0.95, \quad P(B) = 0.97$$

We need to find $P(A \cap B)$. Using the definition of conditional probability, we have -

$$P(A \cap B) = P(B | A) \cdot P(A) = 0.98 \cdot 0.95 = 0.931$$

5. Let the events be defined as follows: A_p : Program is routed to printer A, B_p : Program is routed to printer B, C_p : Program is routed to printer C, A_j : Program jams in printer A, B_j : Program jams in printer B, C_j : Program jams in printer C. We have -

$$P(A_p) = 0.6, \quad P(B_p) = 0.3, \quad P(C_p) = 0.1$$

$$P(A_j) = 0.01, \quad P(B_j) = 0.05, \quad P(C_j) = 0.04$$

We need to find $P(J) = P(A_j \cup B_j \cup C_j)$. Using the law of total probability, we have -

$$P(J) = P(A_j | A_p) \cdot P(A_p) + P(B_j | B_p) \cdot P(B_p) + P(C_j | C_p) \cdot P(C_p)$$

$$= 0.01 \cdot 0.6 + 0.05 \cdot 0.3 + 0.04 \cdot 0.1 = 0.006 + 0.015 + 0.004 = 0.025$$

- (a) We need to find $P(A_j | J)$. Using Bayes' theorem, we have -

$$P(A_j | J) = \frac{P(J | A_j) \cdot P(A_j)}{P(J)} = \frac{1 \cdot P(A_j)}{P(J)}$$

$$= \frac{P(A_j | A_p) \cdot P(A_p)}{P(J)} = \frac{0.01 \cdot 0.6}{0.025} = \frac{0.006}{0.025} = 0.24$$

- (b) We need to find $P(B_j | J)$. Using Bayes' theorem, we have -

$$P(B_j | J) = \frac{P(J | B_j) \cdot P(B_j)}{P(J)} = \frac{1 \cdot P(B_j)}{P(J)}$$

$$= \frac{P(B_j | B_p) \cdot P(B_p)}{P(J)} = \frac{0.05 \cdot 0.3}{0.025} = \frac{0.015}{0.025} = 0.6$$

(c) We need to find $P(C_j | J)$. Using Bayes' theorem, we have -

$$P(C_j | J) = \frac{P(J | C_j) \cdot P(C_j)}{P(J)} = \frac{1 \cdot P(C_j)}{P(J)}$$

$$= \frac{P(C_j | C_p) \cdot P(C_p)}{P(J)} = \frac{0.04 \cdot 0.1}{0.025} = \frac{0.004}{0.025} = 0.16$$

6. Let the events be defined as follows: A : Process shutdown due to equipment failure, B : Process shutdown due to operator Error. We have -

$$P(A) = 0.1, \quad P(B) = 0.4, \quad P(A \cap B) = 0.05$$

(a) We need to find $P(A \cup B)$. Using inclusion-exclusion, we have -

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.1 + 0.4 - 0.05 = 0.45$$

(b) We need to find $P(B \setminus A)$. We have -

$$P(B \setminus A) = P(B) - P(A \cap B) = 0.4 - 0.05 = 0.35$$

(c) We need to find $P(A^C \cap B^C)$. Using De Morgan's law, we have -

$$P(A^C \cap B^C) = 1 - P(A \cup B) = 1 - 0.45 = 0.55$$

(d) We need to find $P(B | A)$. Using the definition of conditional probability, we have -

$$P(B | A) = \frac{P(A \cap B)}{P(A)} = \frac{0.05}{0.1} = 0.5$$

(e) We need to find $P(B | A^C)$. Using the definition of conditional probability, we have -

$$P(B | A^C) = \frac{P(B \setminus A)}{P(A^C)} = \frac{0.35}{0.9} \approx 0.3889$$

7. Let the events be defined as follows: A : The welds are improper. Let B : The machine indicates a defect. We have -

$$P(A) = 0.1, \quad P(B | A^C) = 0.05, \quad P(B | A) = 0.8$$

We need to find $P(A^C | B)$. Using Bayes' theorem, we have -

$$P(A^C | B) = \frac{P(B | A^C) \cdot P(A^C)}{P(B | A^C) \cdot P(A^C) + P(B | A) \cdot P(A)}$$

$$= \frac{0.05 \cdot 0.9}{0.05 \cdot 0.9 + 0.8 \cdot 0.1} = \frac{0.045}{0.045 + 0.08} = \frac{0.045}{0.125} = 0.36$$

8. Let the events be defined as follows: A : First exchange goofed up, B : Second exchange goofed up, C : Third exchange goofed up, E : Bag missing. We have -

$$P(A) = 0.4, \quad P(B) = 0.2, \quad P(C) = 0.1$$

We need to find $P(E)$. We have -

$$P(E) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$

Since the exchanges are independent, we have -

$$P(A \cap B) = P(A) \cdot P(B) = 0.4 \cdot 0.2 = 0.08$$

$$P(A \cap C) = P(A) \cdot P(C) = 0.4 \cdot 0.1 = 0.04$$

$$P(B \cap C) = P(B) \cdot P(C) = 0.2 \cdot 0.1 = 0.02$$

$$P(A \cap B \cap C) = P(A) \cdot P(B) \cdot P(C) = 0.4 \cdot 0.2 \cdot 0.1 = 0.008$$

Thus,

$$P(E) = 0.4 + 0.2 + 0.1 - 0.08 - 0.04 - 0.02 + 0.008 = 0.558$$

Now we need to find $P(A | E)$. Using Bayes' theorem, we have -

$$P(A | E) = \frac{P(E | A) \cdot P(A)}{P(E)}$$

Since if the first exchange goofed up, the bag is definitely missing, we have $P(E | A) = 1$. Thus,

$$P(A | E) = \frac{1 \cdot 0.4}{0.558} \approx 0.716$$

9. Let the cups and saucers be defined as follows: There are two colors — red and blue. Let the cups be R_1, R_2, B_1, B_2 and the saucers be R'_1, R'_2, B'_1, B'_2 . Each cup is placed randomly on one saucer. The total number of possible arrangements is -

$$\text{Total arrangements} = 4! = 24$$

We need to find the probability that no cup is placed on a saucer of the same color. This means that the red cups must be placed on blue saucers and the blue cups on red saucers. Hence,

$$\text{Number of ways to place red cups on blue saucers} = 2! = 2,$$

$$\text{Number of ways to place blue cups on red saucers} = 2! = 2.$$

Therefore, the total number of favorable arrangements is -

$$\text{Favorable arrangements} = 2! \times 2! = 4.$$

Thus, the required probability is -

$$P(\text{no cup on same color}) = \frac{\text{favorable arrangements}}{\text{total arrangements}} = \frac{4}{24} = \frac{1}{6}$$

$$P = \frac{1}{6}$$

10. Let the events be defined as follows: A : Sum on the two die is 7, B : Sum on the two die is 11, L_1 : Sum on the die is 3, L_2 : Sum on the die is 2, L_3 : Sum on the die is 12, l_f : Second or further sum is 7, j : The sum is any number other than ones which win or lose. We have -

$$P(A) = \frac{6}{36}, \quad P(B) = \frac{2}{36}, \quad P(L_1) = \frac{2}{36}, \quad P(L_2) = \frac{1}{36}, \quad P(L_3) = \frac{1}{36}$$

Case 1: Win on First Throw:

We need to find $P(A \cup B)$. Using inclusion-exclusion, we have -

$$P(A \cup B) = P(A) + P(B) = \frac{6}{36} + \frac{2}{36} = \frac{8}{36} = \frac{2}{9}$$

Case 2: Win on Subsequent Throws:

We need to find $P(\text{Win}) = P(j \cup B \cup L_1^C \cup L_2^C \cup L_3^C \cup l_f^C)$. Using inclusion-exclusion, we have -

$$P(\text{Win}) = P(j) + P(B) + P(L_1^C) + P(L_2^C) + P(L_3^C) + P(l_f^C) - (\text{intersections})$$

However, calculating all intersections is complex. Instead, we can use the complementary probability approach. The probability of losing on subsequent throws is the sum of probabilities of losing on each losing condition:

$$P(\text{Lose}) = P(L_1) + P(L_2) + P(L_3) + P(l_f)$$

$$= \frac{2}{36} + \frac{1}{36} + \frac{1}{36} + \frac{6}{36} = \frac{10}{36} = \frac{5}{18}$$

Thus, the probability of winning on subsequent throws is -

$$P(\text{Win on subsequent throws}) = 1 - P(\text{Lose}) = 1 - \frac{5}{18} = \frac{13}{18}$$

Therefore, the total probability of winning the game is -

$$P(\text{Total Win}) = P(\text{Win on First Throw}) + P(\text{Win on subsequent throws}) = \frac{2}{9} + \frac{13}{18} = \frac{4}{18} + \frac{13}{18} = \frac{17}{18}$$

Assignment 4

Prob Stats 1: Assignment Solutions

Eklavya Raman - 24MS024

1. Let the events be defined as follows: A_n : A family has n children, B : A family has a boy, C : A family has a girl. We have -

$$P(A_n) = kp^n, \quad n = 0, 1, 2, \dots$$

where k is a normalizing constant. Also, we have -

$$P(B) = P(C) = 0.5$$

- (a) We need to find the probability that a family has at least 1 boy. We have -

$$\sum_{i=1}^n P(B | A_i) \cdot P(A_i) = \sum_{i=1}^n \left(1 - \left(\frac{1}{2}\right)^i\right) \cdot kp^i$$

Thus, the probability that a family has at least 1 boy is -

$$P(\text{at least 1 boy}) = \sum_{i=1}^n \left(1 - \left(\frac{1}{2}\right)^i\right) \cdot kp^i$$

Let $n \rightarrow \infty$. We have -

$$P(\text{at least 1 boy}) = \sum_{i=1}^{\infty} \left(1 - \left(\frac{1}{2}\right)^i\right) \cdot kp^i$$

We can calculate this as follows -

$$P(\text{at least 1 boy}) = k \left(\frac{p}{1-p} - \frac{p/2}{1-p/2} \right) = k \left(\frac{p}{1-p} - \frac{p}{2-p} \right)$$

Let us now find the value of k . We have -

$$\sum_{i=0}^{\infty} P(A_i) = \sum_{i=0}^{\infty} kp^i = k \cdot \frac{1}{1-p} = 1$$

Thus, we have -

$$k = 1 - p$$

Therefore, the probability that a family has at least 1 boy is -

$$P(\text{at least 1 boy}) = (1-p) \left(\frac{p}{1-p} - \frac{p}{2-p} \right) = p - \frac{p(1-p)}{2-p} = p \left(1 - \frac{1-p}{2-p} \right) = p \frac{(2-p) - (1-p)}{2-p} = \frac{p}{2-p}$$

- (b) We need to find the probability that if a family has at least one boy, what is the probability that the family has two or more boys. We have -

$$P(2 \text{ or more boys} | \text{at least 1 boy}) = \frac{P(2 \text{ or more boys} \cap \text{at least 1 boy})}{P(\text{at least 1 boy})} = \frac{P(2 \text{ or more boys})}{P(\text{at least 1 boy})}$$

We can calculate $P(2 \text{ or more boys})$ as follows -

$$P(2 \text{ or more boys}) = \sum_{i=2}^{\infty} P(2 \text{ or more boys} \mid A_i) \cdot P(A_i) = \sum_{i=2}^{\infty} \left(1 - \left(\frac{1}{2} \right)^i - i \left(\frac{1}{2} \right)^i \right) \cdot kp^i$$

Thus, we have -

$$P(2 \text{ or more boys}) = \sum_{i=2}^{\infty} \left(1 - \left(\frac{1}{2} \right)^i - i \left(\frac{1}{2} \right)^i \right) \cdot kp^i$$

Let $n \rightarrow \infty$. We have -

$$P(2 \text{ or more boys}) = \sum_{i=2}^{\infty} \left(1 - \left(\frac{1}{2} \right)^i - i \left(\frac{1}{2} \right)^i \right) \cdot kp^i$$

We can calculate this as follows -

$$P(2 \text{ or more boys}) = k \left(\frac{p}{1-p} - \frac{p/2}{1-p/2} - \frac{(p/2)^2}{(1-p/2)^2} \right) = k \left(\frac{p}{1-p} - \frac{p}{2-p} - \frac{p^2}{(2-p)^2} \right)$$

Therefore, the probability that if a family has at least one boy, what is the probability that the family has two or more boys is -

$$\frac{(1-p) \left(\frac{p}{1-p} - \frac{p}{2-p} - \frac{p^2}{(2-p)^2} \right)}{\frac{p}{2-p}} = \frac{(2-p) - (1-p) - \frac{p(1-p)}{2-p}}{(2-p)} = \frac{1 - \frac{p(1-p)}{2-p}}{(2-p)} = 1 - \frac{p(1-p)}{(2-p)^2}$$

2. Let the events be defined as follows: A : Computer failure due to overload, B : Computer failure due to software error. We have -

$$P(A) = 0.75, \quad P(B) = 0.15, \quad P(A \cup B) = 0.85$$

We need to find $P(A \cap B)$. Using inclusion-exclusion, we have -

$$P(A \cap B) = P(A) + P(B) - P(A \cup B) = 0.75 + 0.15 - 0.85 = 0.05$$

Now, we need to find $P(B \cap A^C)$. We have -

$$P(B \cap A^C) = P(B) - P(A \cap B) = 0.15 - 0.05 = 0.1$$

3. The pdf is given as -

x	0	1	2	3	4	5	6	7
$P(X = x)$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2 + k$

- (a) We know that the sum of all probabilities must equal 1. Thus, we have -

$$0 + k + 2k + 2k + 3k + k^2 + 2k^2 + 7k^2 + k = 1$$

$$8k + 10k^2 = 1$$

$$10k^2 + 8k - 1 = 0$$

Solving this quadratic equation, we get -

$$k = \frac{-8 \pm \sqrt{64 + 40}}{20} = \frac{-8 \pm \sqrt{104}}{20} = \frac{-8 \pm 2\sqrt{26}}{20} = \frac{-4 \pm \sqrt{26}}{10}$$

Since k must be positive, we take the positive root -

$$k = \frac{-4 + \sqrt{26}}{10}$$

(b) We need to find $P(X < 6)$, $P(X \geq 6)$ and $P(0 < X < 5)$. We have -

$$P(X < 6) = P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4) + P(X = 5)$$

$$= 0 + k + 2k + 2k + 3k + k^2 = 8k + k^2 = 8 \left(\frac{-4 + \sqrt{26}}{10} \right) + \left(\frac{-4 + \sqrt{26}}{10} \right)^2$$

$$P(X \geq 6) = P(X = 6) + P(X = 7)$$

$$= 2k^2 + 7k^2 + k = 9k^2 + k = 9 \left(\frac{-4 + \sqrt{26}}{10} \right)^2 + \left(\frac{-4 + \sqrt{26}}{10} \right)$$

$$P(0 < X < 5) = P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4)$$

$$= k + 2k + 2k + 3k = 8k = 8 \left(\frac{-4 + \sqrt{26}}{10} \right)$$

(c) If $P(X \leq a) > 0.5$, find the minimum value of a . We have -

$$P(X \leq a) = P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4) + P(X = 5) + P(X = 6) + P(X = 7)$$

We need to find the minimum value of a such that this sum is greater than 0.5. Calculating the cumulative probabilities, we find that -

$$P(X \leq 4) = 8k$$

$$P(X \leq 5) = 8k + k^2$$

$$P(X \leq 6) = 8k + k^2 + 2k^2 = 8k + 3k^2$$

$$P(X \leq 7) = 8k + 3k^2 + 7k^2 + k = 9k + 10k^2 = 1$$

We need to check when these cumulative probabilities exceed 0.5. After calculation, we find that -

$$a = 5$$

(d) Determine the Distribution Function $F(x)$. The distribution function is given by -

$$F(x) = \begin{cases} 0 & x < 0 \\ 0 & 0 \leq x < 1 \\ k & 1 \leq x < 2 \\ 3k & 2 \leq x < 3 \\ 5k & 3 \leq x < 4 \\ 8k & 4 \leq x < 5 \\ 8k + k^2 & 5 \leq x < 6 \\ 8k + 3k^2 & 6 \leq x < 7 \\ 1 & x \geq 7 \end{cases}$$

$$\text{where } k = \frac{-4 + \sqrt{26}}{10}.$$

4. We have

$$P(X = x) = \begin{cases} \frac{x}{15}, & x = 1, 2, 3, 4, 5 \\ 0, & \text{otherwise} \end{cases}$$

(a) We need to find $P(X = 1 \text{ or } X = 2)$. We have -

$$P(X = 1 \text{ or } X = 2) = P(X = 1) + P(X = 2) = \frac{1}{15} + \frac{2}{15} = \frac{3}{15} = \frac{1}{5}$$

(b) We need to find $P[1/2 < X < 5/2 \mid X \geq 1]$. We have -

$$P\left(\frac{1}{2} < X < \frac{5}{2} \mid X \geq 1\right) = \frac{P(1 < X < \frac{5}{2})}{P(X \geq 1)} = \frac{P(X = 2)}{1} = \frac{2}{15}$$

$$\boxed{P\left(\frac{1}{2} < X < \frac{5}{2} \mid X \geq 1\right) = \frac{2}{15}}$$

5. Let the events be defined as follows: X : Sum of the numbers on the two dice. We need to find the pmf of X . The possible sums when rolling two dice are 2 to 12. The pmf is given by -

$$P(X = x) = \begin{cases} \frac{1}{36}, & x = 2 \\ \frac{2}{36}, & x = 3 \\ \frac{3}{36}, & x = 4 \\ \frac{4}{36}, & x = 5 \\ \frac{5}{36}, & x = 6 \\ \frac{6}{36}, & x = 7 \\ \frac{5}{36}, & x = 8 \\ \frac{4}{36}, & x = 9 \\ \frac{3}{36}, & x = 10 \\ \frac{2}{36}, & x = 11 \\ \frac{1}{36}, & x = 12 \\ 0, & \text{otherwise} \end{cases}$$

We also need to find the distribution function $F(x)$. The distribution function is given by -

$$F(x) = \begin{cases} 0 & x < 2 \\ \frac{1}{36} & 2 \leq x < 3 \\ \frac{3}{36} & 3 \leq x < 4 \\ \frac{6}{36} & 4 \leq x < 5 \\ \frac{10}{36} & 5 \leq x < 6 \\ \frac{15}{36} & 6 \leq x < 7 \\ \frac{21}{36} & 7 \leq x < 8 \\ \frac{26}{36} & 8 \leq x < 9 \\ \frac{30}{36} & 9 \leq x < 10 \\ \frac{33}{36} & 10 \leq x < 11 \\ \frac{35}{36} & 11 \leq x < 12 \\ 1 & x \geq 12 \end{cases}$$

6. We have the following pdf -

$$f(x) = \begin{cases} ax, & 0 \leq x \leq 1 \\ a, & 1 \leq x \leq 2 \\ 3a - ax, & 2 \leq x \leq 3 \\ 0, & \text{otherwise} \end{cases}$$

We need to find the value of a . We know that the integral of the pdf over its entire range must equal 1. Thus, we have -

$$\int_0^1 ax \, dx + \int_1^2 a \, dx + \int_2^3 (3a - ax) \, dx = 1$$

Calculating each integral, we get -

$$\frac{a}{2} + a + \left(3a - \frac{3a}{2}\right) = 1$$

$$\frac{a}{2} + a + \frac{3a}{2} = 1$$

$$3a = 1$$

$$\boxed{a = \frac{1}{3}}$$

We now compute $P(X \leq 1.5)$. We have -

$$P(X \leq 1.5) = \int_0^1 ax \, dx + \int_1^{1.5} a \, dx$$

Calculating each integral, we get -

$$\frac{a}{2} + a(1.5 - 1) = \frac{a}{2} + \frac{a}{2} = a$$

$$\boxed{P(X \leq 1.5) = \frac{1}{3}}$$

7. Let the life (in hours) of the part be the r.v. X with p.d.f.

$$f(x) = \begin{cases} \frac{100}{x^2}, & x > 100, \\ 0, & \text{elsewhere.} \end{cases}$$

Note that this is a valid pdf since

$$\int_{100}^{\infty} \frac{100}{x^2} \, dx = 100 \left[-\frac{1}{x} \right]_{100}^{\infty} = 100 \cdot \frac{1}{100} = 1.$$

For $t > 100$ the survival function is

$$P(X > t) = \int_t^{\infty} \frac{100}{x^2} \, dx = \frac{100}{t},$$

and the c.d.f. is

$$F(t) = P(X \leq t) = 1 - \frac{100}{t} \quad (t > 100).$$

(a) **Probability all three tubes must be replaced during first 150 hours.**

For one tube,

$$P(X \leq 150) = F(150) = 1 - \frac{100}{150} = 1 - \frac{2}{3} = \frac{1}{3}.$$

For three independent tubes the probability that all three fail by 150 hours is

$$\boxed{\left(\frac{1}{3}\right)^3 = \frac{1}{27}}.$$

- (b) **Probability none of three tubes must be replaced during first 150 hours.**

For one tube,

$$P(X > 150) = \frac{100}{150} = \frac{2}{3}.$$

Hence for three independent tubes

$$\left(\frac{2}{3}\right)^3 = \frac{8}{27}.$$

- (c) **Probability a tube lasts less than 200 hours given it is working after 150 hours.**

We want $P(X < 200 \mid X > 150)$. Compute

$$P(150 < X < 200) = F(200) - F(150) = \left(1 - \frac{100}{200}\right) - \left(1 - \frac{100}{150}\right) = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}.$$

Since $P(X > 150) = \frac{2}{3}$,

$$P(X < 200 \mid X > 150) = \frac{P(150 < X < 200)}{P(X > 150)} = \frac{\frac{1}{6}}{\frac{2}{3}} = \frac{1}{4}.$$

Hence

$$P(X < 200 \mid X > 150) = \frac{1}{4}.$$

(Equivalently $1 - \frac{S(200)}{S(150)} = 1 - \frac{(100/200)}{(100/150)} = 1 - \frac{1/2}{2/3} = 1 - \frac{3}{4} = \frac{1}{4}$.)

- (d) **Maximum number of tubes n so that $P(\text{all still functioning after 150 h}) \geq 0.5$.**

For n independent tubes the probability all survive 150 hours is

$$(P(X > 150))^n = \left(\frac{2}{3}\right)^n.$$

We require $\left(\frac{2}{3}\right)^n \geq \frac{1}{2}$. Taking logarithms,

$$n \leq \frac{\ln(1/2)}{\ln(2/3)} \approx \frac{-0.693147}{-0.405465} \approx 1.7095.$$

Since n must be an integer, the largest integer n satisfying the inequality is $n = 1$. (Check: for $n = 1$, $2/3 \geq 1/2$; for $n = 2$, $(2/3)^2 = 4/9 \approx 0.444 < 1/2$.)

Therefore the maximum number is

$$n_{\max} = 1.$$

8. We need to count the number of α particles given off in 2 seconds by a radioactive source. Let the random variable X denote this count. We know that on average 3.2 such particles are given off. We know that X follows a Poisson distribution with parameter $\lambda = 3.2$. Thus, the pmf of X is given by -

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!} = \frac{e^{-3.2} (3.2)^x}{x!}, \quad x = 0, 1, 2, \dots$$

We need to find the probability that no more than 2 particles are given off in 2 seconds. Thus, we have -

$$P(X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2)$$

$$= \frac{e^{-3.2} (3.2)^0}{0!} + \frac{e^{-3.2} (3.2)^1}{1!} + \frac{e^{-3.2} (3.2)^2}{2!} = e^{-3.2} \left(1 + 3.2 + \frac{(3.2)^2}{2}\right) \approx 0.3799$$

9. We have the following pdf with x representing the number of minutes one has to wait for a bus at a downtown bus stop-

$$f(x) = \begin{cases} 0, & x < 0 \\ \frac{1}{9}(x+1), & 0 \leq x < 1 \\ \frac{4}{9}(x-1/2), & 1 \leq x < 1.5 \\ \frac{4}{9}(5/2-x), & 1.5 \leq x < 2 \\ \frac{1}{9}(4-x), & 2 \leq x < 3 \\ \frac{1}{9}, & 3 \leq x < 6 \\ \frac{1}{9}, & 6 \leq x \end{cases}$$

Let the cases be defined as follows: A : One waits between 0 and 2 minutes inclusive, B : One waits between 1 and 3 minutes inclusive. We have -

- (a) We need to find $P(A)$. We have -

$$P(A) = \int_0^1 \frac{1}{9}(x+1) dx + \int_1^{1.5} \frac{4}{9}(x-1/2) dx + \int_{1.5}^2 \frac{4}{9}(5/2-x) dx$$

Calculating each integral, we get -

$$P(A) = \frac{5}{18} + \frac{5}{72} + \frac{5}{72} = \frac{25}{72}$$

- (b) We need to find $P(B)$. We have -

$$P(B) = \int_1^{1.5} \frac{4}{9}(x-1/2) dx + \int_{1.5}^2 \frac{4}{9}(5/2-x) dx + \int_2^3 \frac{1}{9}(4-x) dx$$

Calculating each integral, we get -

$$P(B) = \frac{5}{72} + \frac{5}{72} + \frac{5}{18} = \frac{25}{72}$$

- (c) We need to find $P(B | A)$. We have -

$$P(B | A) = \frac{P(B \cap A)}{P(A)}$$

Since $B \cap A$ is the event that one waits between 1 and 2 minutes inclusive, we have -

$$P(B \cap A) = \int_1^{1.5} \frac{4}{9}(x-1/2) dx + \int_{1.5}^2 \frac{4}{9}(5/2-x) dx$$

Calculating each integral, we get -

$$P(B \cap A) = \frac{5}{72} + \frac{5}{72} = \frac{5}{36}$$

Thus, we have -

$$P(B | A) = \frac{\frac{5}{36}}{\frac{25}{72}} = \frac{5}{36} \cdot \frac{72}{25} = \frac{10}{25} = \frac{2}{5}$$

(d) We need to find $P(A^C | B^C)$. We have -

$$P(A^C | B^C) = \frac{P(A^C \cap B^C)}{P(B^C)} = \frac{P((A \cup B)^C)}{1 - P(B)}$$

Since $A \cup B$ is the event that one waits between 0 and 3 minutes inclusive, we have -

$$P(A \cup B) = \int_0^1 \frac{1}{9}(x+1) dx + \int_1^{1.5} \frac{4}{9}(x-1/2) dx + \int_{1.5}^2 \frac{4}{9}(5/2-x) dx + \int_2^3 \frac{1}{9}(4-x) dx$$

Calculating each integral, we get -

$$P(A \cup B) = \frac{5}{18} + \frac{5}{72} + \frac{5}{72} + \frac{5}{18} = \frac{50}{72} = \frac{25}{36}$$

Thus, we have -

$$P(A^C | B^C) = \frac{1 - \frac{25}{36}}{1 - \frac{25}{72}} = \frac{\frac{11}{36}}{\frac{47}{72}} = \frac{11}{36} \cdot \frac{72}{47} = \frac{22}{47}$$

10. We need to verify whether the following function is a valid cdf -

$$F(x) = \begin{cases} 0, & x < 0 \\ 0.1, & 0 \leq x < 0.1 \\ x, & 0.1 \leq x < 0.2 \\ x^2 + 0.2, & 0.2 \leq x < 0.3 \\ 0.3, & 0.3 \leq x < 0.4 \\ x^3 + x^2 + 0.076, & 0.4 \leq x < 0.5 \\ 0.5, & 0.5 \leq x < 0.6 \\ x - 0.1, & 0.6 \leq x < 1 \\ 1, & x \geq 1 \end{cases}$$

To verify whether $F(x)$ is a valid cdf, we need to check the following conditions:

- (a) $F(x)$ is non-decreasing.
- (b) $\lim_{x \rightarrow -\infty} F(x) = 0$ and $\lim_{x \rightarrow \infty} F(x) = 1$.
- (c) $F(x)$ is right-continuous.

(a) Checking non-decreasing property:

- From 0 to 0.1, $F(x) = 0.1$ (constant).
- From 0.1 to 0.2, $F(x) = x$ (increasing).
- From 0.2 to 0.3, $F(x) = x^2 + 0.2$ (increasing).
- From 0.3 to 0.4, $F(x) = 0.3$ (constant).
- From 0.4 to 0.5, $F(x) = x^3 + x^2 + 0.076$ (increasing).
- From 0.5 to 0.6, $F(x) = 0.5$ (constant).
- From 0.6 to 1, $F(x) = x - 0.1$ (increasing).

Thus, $F(x)$ is non-decreasing.

(b) Checking limits:

$$\lim_{x \rightarrow -\infty} F(x) = 0$$

$$\lim_{x \rightarrow \infty} F(x) = 1$$

(c) Checking right-continuity: At each point of discontinuity, we check the right-hand limit:

- At $x = 0$: $\lim_{x \rightarrow 0^+} F(x) = 0.1 = F(0)$.
- At $x = 0.1$: $\lim_{x \rightarrow 0.1^+} F(x) = 0.1 = F(0.1)$.
- At $x = 0.2$: $\lim_{x \rightarrow 0.2^+} F(x) = 0.24 = F(0.2)$.

- At $x = 0.3$: $\lim_{x \rightarrow 0.3^+} F(x) = 0.3 = F(0.3)$.
- At $x = 0.4$: $\lim_{x \rightarrow 0.4^+} F(x) = 0.5 = F(0.4)$.
- At $x = 0.5$: $\lim_{x \rightarrow 0.5^+} F(x) = 0.5 = F(0.5)$.
- At $x = 0.6$: $\lim_{x \rightarrow 0.6^+} F(x) = 0.5 = F(0.6)$.

Thus, $F(x)$ is right-continuous.

Since all three conditions are satisfied, we conclude that $F(x)$ is a valid cdf.

Assignment 5

Prob Stats 1: Assignment Solutions

Eklavya Raman - 24MS024

1. Let X have pdf

$$f(x) = \begin{cases} kx^2(1-x^3), & 0 \leq x \leq 1, \\ 0, & \text{elsewhere.} \end{cases}$$

Normalizing constant k .

Compute $\int_0^1 x^2(1-x^3) dx = \int_0^1 (x^2 - x^5) dx = \left[\frac{x^3}{3} - \frac{x^6}{6} \right]_0^1 = \frac{1}{3} - \frac{1}{6} = \frac{1}{6}$. Hence $k \cdot \frac{1}{6} = 1 \Rightarrow \boxed{k = 6}$.

With $k = 6$ the moments are (all integrals over $[0, 1]$):

$$E[X] = 6 \int_0^1 (x^3 - x^6) dx = 6 \left(\frac{1}{4} - \frac{1}{7} \right) = 6 \cdot \frac{3}{28} = \frac{18}{28} = \boxed{\frac{9}{14} \approx 0.642857},$$

$$E[X^2] = 6 \int_0^1 (x^4 - x^7) dx = 6 \left(\frac{1}{5} - \frac{1}{8} \right) = 6 \cdot \frac{3}{40} = \frac{18}{40} = \boxed{\frac{9}{20} = 0.45},$$

$$E[X^3] = 6 \int_0^1 (x^5 - x^8) dx = 6 \left(\frac{1}{6} - \frac{1}{9} \right) = 6 \cdot \frac{1}{18} = \boxed{\frac{1}{3} \approx 0.333333},$$

$$E[X^4] = 6 \int_0^1 (x^6 - x^9) dx = 6 \left(\frac{1}{7} - \frac{1}{10} \right) = 6 \cdot \frac{3}{70} = \boxed{\frac{9}{35} \approx 0.257143}.$$

Variance.

$$\text{Var}(X) = E[X^2] - (E[X])^2 = \frac{9}{20} - \left(\frac{9}{14} \right)^2 = \frac{9}{245} \approx 0.0367347.$$

$$\Rightarrow \text{Var}(X) = \frac{9}{245} \approx 0.03673.$$

Central moments, skewness and kurtosis.

Compute the third central moment

$$\mu_3 = E[(X - \mu)^3] = E[X^3] - 3\mu E[X^2] + 2\mu^3 = -\frac{131}{41160} \approx -0.00318270,$$

and the fourth central moment

$$\mu_4 = E[(X - \mu)^4] = E[X^4] - 4\mu E[X^3] + 6\mu^2 E[X^2] - 3\mu^4 = \frac{663}{192080} \approx 0.00345169.$$

Skewness

$$\gamma_1 = \frac{\mu_3}{\sigma^3} \approx \frac{-0.00318270165}{(0.0367346939)^{3/2}} \approx \boxed{-0.4520}.$$

Kurtosis

$$\beta_2 = \frac{\mu_4}{\sigma^4} \approx \frac{0.00345169}{(0.0367346939)^2} \approx \boxed{2.5565}.$$

2. We have the following pdf -

x	-3	6	9
$P(X = x)$	$\frac{1}{16}$	$\frac{1}{2}$	$\frac{1}{3}$

We need to calculate $E(X)$, $V(X)$ and $E(2X + 1)$. We have -

(a) Calculating $E(X)$, we have -

$$E(X) = \sum x \cdot P(X = x) = -3 \cdot \frac{1}{16} + 6 \cdot \frac{1}{2} + 9 \cdot \frac{1}{3}$$

$$= -\frac{3}{16} + 3 + 3 = \frac{93}{16} = 5.8125$$

(b) Calculating $V(X)$, we have -

$$V(X) = E(X^2) - (E(X))^2$$

where

$$E(X^2) = \sum x^2 \cdot P(X = x) = (-3)^2 \cdot \frac{1}{16} + 6^2 \cdot \frac{1}{2} + 9^2 \cdot \frac{1}{3}$$

$$= \frac{9}{16} + 18 + 27 = \frac{783}{16} = 48.9375$$

Thus, we have -

$$V(X) = \frac{783}{16} - \left(\frac{93}{16}\right)^2 = \frac{783}{16} - \frac{8649}{256} = \frac{783 \cdot 16 - 8649}{256} = \frac{12528 - 8649}{256} = \frac{3879}{256} \approx 15.1445$$

(c) Calculating $E(2X + 1)$, we have -

$$E(2X + 1) = 2E(X) + 1 = 2 \cdot \frac{93}{16} + 1$$

$$= \frac{186}{16} + \frac{16}{16} = \frac{202}{16} = \frac{101}{8} = 12.625$$

3. We have a random variable X with the following distribution -

$$P(X = 0) = P(X = 2) = p, \quad P(X = 1) = 1 - 2p, \quad 0 \leq p \leq 0.5$$

We need to find the value of p such that the variance of X is maximized. We have -

$$E(X) = 0 \cdot p + 1 \cdot (1 - 2p) + 2 \cdot p = 1$$

$$E(X^2) = 0^2 \cdot p + 1^2 \cdot (1 - 2p) + 2^2 \cdot p = 4p + 1 - 2p = 1 + 2p$$

Thus, we have -

$$V(X) = E(X^2) - (E(X))^2 = (1 + 2p) - 1^2 = 2p$$

To maximize $V(X)$, we need to maximize $2p$. Since p is bounded between 0 and 0.5, the maximum value of p is 0.5. Thus, we have -

$$p = 0.5$$

4. Let the events be defined as follows: +: Die shows 4, 5, 6, -: Die shows 1, 2, 3, X_i : Neighboring signs are different after $(i + 1)$ rolls. We need to find the probability that neighboring signs are different when 6 dice are rolled.

We note that the signs are bernoulli random variables with parameter $p = \frac{1}{2}$ since there are equal chances of getting a + or - sign. We have -

$$P(X = 1) = P(+)P(-)P(+) + P(-)P(+)P(-) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

$$P(X = 0) = 1 - P(X = 1) = 1 - \frac{1}{4} = \frac{3}{4}$$

We need the variance and expectation of X . We have -

$$E(X) = \sum x \cdot P(X = x) = 0 \cdot \frac{3}{4} + 1 \cdot \frac{1}{4} = \frac{1}{4}$$

$$V(X) = E(X^2) - (E(X))^2 = \left(0^2 \cdot \frac{3}{4} + 1^2 \cdot \frac{1}{4}\right) - \left(\frac{1}{4}\right)^2 = \frac{1}{4} - \frac{1}{16} = \frac{3}{16}$$

5. We have X is a nonnegative continuous random variable with cdf $F(x)$. We need to show that $E(X) = \int_0^\infty (1 - F(x)) dx$.

Proof. We know that the expectation of a nonnegative continuous random variable is given by -

$$E(X) = \int_0^\infty x f(x) dx$$

where $f(x)$ is the pdf of X . We also know that the cdf $F(x)$ is related to the pdf $f(x)$ by -

$$F(x) = \int_0^x f(t) dt$$

Thus, we have -

$$1 - F(x) = 1 - \int_0^x f(t) dt = \int_x^\infty f(t) dt$$

Let us now integrate $1 - F(x)$ from 0 to ∞ :

$$\int_0^\infty (1 - F(x)) dx = \int_0^\infty \int_x^\infty f(t) dt dx$$

We can change the order of integration, and we have -

$$\int_0^\infty (1 - F(x)) dx = \int_0^\infty \int_0^t f(t) dx dt = \int_0^\infty t f(t) dt$$

Thus, we have shown that -

$$E(X) = \int_0^\infty (1 - F(x)) dx$$

□

6. This problem can be modeled using a binomial distribution where each word has a probability of 0.001 of having a misprint. We have the following parameters: $N = 350$ (number of trials/words), $p = 0.001$ (probability of success/misprint on each word), $q = 1 - p = 0.999$ (probability of no misprint on each word). We need to find the probability that at least one word has a misprint. Thus, we have -

$$P(\text{at least one misprint}) = 1 - P(\text{no misprints})$$

$$= 1 - \binom{350}{0} p^0 q^{350} = 1 - (1)(1)(0.999^{350})$$

$$= 1 - 0.7047 \approx 0.2953$$

7. We have that the owner of a small hotel with 5 rooms has 3 television sets. It is expected that half of the guests will want a TV. We have full occupancy of the hotel.

- (a) The probability that there will be more requests than TV's available can be modeled using a binomial distribution where each guest has a probability of 0.5 of requesting a TV. We have the following parameters: $N = 5$ (number of trials/guests), $p = 0.5$ (probability of success/requesting a TV), $q = 1 - p = 0.5$ (probability of no request for a TV). We need to find the probability that more than 3 guests request a TV. Thus, we have -

$$P(X > 3) = P(X = 4) + P(X = 5)$$

$$= \binom{5}{4} p^4 q^1 + \binom{5}{5} p^5 q^0$$

$$= 5 \cdot (0.5)^4 \cdot (0.5)^1 + 1 \cdot (0.5)^5 \cdot (1) = \frac{5}{32} + \frac{1}{32} = \frac{6}{32} = \frac{3}{16} = 0.1875$$

(b) We need to find the probability that a request was made and a TV was available. We have -

$$\begin{aligned}
 P(\text{request made and TV available}) &= P(X \leq 3) \\
 &= P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3) \\
 &= \binom{5}{0} p^0 q^5 + \binom{5}{1} p^1 q^4 + \binom{5}{2} p^2 q^3 + \binom{5}{3} p^3 q^2 \\
 &= 1 \cdot (1) \cdot (0.5)^5 + 5 \cdot (0.5)^1 \cdot (0.5)^4 + 10 \cdot (0.5)^2 \cdot (0.5)^3 + 10 \cdot (0.5)^3 \cdot (0.5)^2 \\
 &= \frac{1}{32} + \frac{5}{32} + \frac{10}{32} + \frac{10}{32} = \frac{26}{32} = \frac{13}{16} = 0.8125
 \end{aligned}$$

(c) The cost to operate a TV for one day is C . The expected cost for the hotel owner can be calculated as follows:

$$E(\text{cost}) = C \cdot E(X)$$

where $E(X)$ is the expected number of guests requesting a TV. We have -

$$E(X) = N \cdot p = 5 \cdot 0.5 = 2.5$$

Thus, we have -

$$E(\text{cost}) = C \cdot 2.5 = 2.5C$$

Thus, the rent R that the owner should charge per rented set to break even is given by -

$$R \cdot 3 = E(\text{cost})$$

$$R = \frac{E(\text{cost})}{3} = \frac{2.5C}{3} = \frac{5C}{6}$$

8. We have $P(X = i) = k \frac{1}{i(i+1)}$ for $i \in \mathbb{N}$. First, we determine the value of k by using the fact that the sum of all probabilities must equal 1. Thus, we have -

$$\begin{aligned}
 \sum_{i=1}^{\infty} P(X = i) &= 1 \\
 \sum_{i=1}^{\infty} k \frac{1}{i(i+1)} &= 1
 \end{aligned}$$

We can simplify the summation using partial fraction decomposition:

$$\frac{1}{i(i+1)} = \frac{1}{i} - \frac{1}{i+1}$$

Thus, we have -

$$k \sum_{i=1}^{\infty} \left(\frac{1}{i} - \frac{1}{i+1} \right) = 1$$

This is a telescoping series, and we have -

$$k \left(1 - \lim_{n \rightarrow \infty} \frac{1}{n+1} \right) = 1$$

$$k \cdot 1 = 1$$

$$k = 1$$

Therefore the pmf of X is given by -

$$P(X = i) = \frac{1}{i(i+1)}, \quad i \in \mathbb{N}$$

We need to calculate $P(3 \leq X < 6)$. Thus, we have -

$$\begin{aligned}
 P(3 \leq X < 6) &= P(X = 3) + P(X = 4) + P(X = 5) \\
 &= \frac{1}{3 \cdot 4} + \frac{1}{4 \cdot 5} + \frac{1}{5 \cdot 6} \\
 &= \frac{1}{12} + \frac{1}{20} + \frac{1}{30} = \frac{5}{60} + \frac{3}{60} + \frac{2}{60} = \frac{10}{60} = \frac{1}{6}
 \end{aligned}$$

9. Let the events be defined as follows: A_x : Passenger x does not show up for the flight. We have -

$$P(A_x) = 0.05, \forall x, \quad P(A_x^C) = 0.95, \forall x$$

Airline A sells 63 tickets for a 60-seat airplane. We need to find the probability that more than 60 passengers show up for the flight. Let X be the random variable denoting the number of passengers who show up for the flight. We have -

$$X \sim \text{Binomial}(n = 63, p = 0.95)$$

We need to find $P(X > 60)$. Thus, we have -

$$\begin{aligned}
 P(X > 60) &= P(X = 61) + P(X = 62) + P(X = 63) \\
 &= \binom{63}{61} (0.95)^{61} (0.05)^2 + \binom{63}{62} (0.95)^{62} (0.05)^1 + \binom{63}{63} (0.95)^{63} (0.05)^0 \\
 &= 1953 \cdot (0.95)^{61} \cdot (0.05)^2 + 63 \cdot (0.95)^{62} \cdot (0.05)^1 + 1 \cdot (0.95)^{63} \cdot (1) \approx 0.2271
 \end{aligned}$$

Airline B sells 126 tickets for a 120-seat airplane. We need to find the probability that more than 120 passengers show up for the flight. Let Y be the random variable denoting the number of passengers who show up for the flight. We have -

$$Y \sim \text{Binomial}(n = 126, p = 0.95)$$

We need to find $P(Y > 120)$. Thus, we have -

$$P(Y > 120) = P(Y = 121) + P(Y = 122) + P(Y = 123) + P(Y = 124) + P(Y = 125) + P(Y = 126)$$

$$= \sum_{k=121}^{126} \binom{126}{k} (0.95)^k (0.05)^{126-k} \approx 0.4593$$

Therefore, Airline B is more likely to have a problem with overbooking as the probability of more than 120 passengers showing up is higher than the probability of more than 60 passengers showing up for Airline A.

10. The source emits on average 0.5 particles per second. We need to find the probability that more than 3 particles are emitted in a 5-second interval. This can be modeled using a Poisson distribution with parameter $\lambda = 0.5 \times 5 = 2.5$. Thus, we have -

$$\begin{aligned}
 P(X > 3) &= 1 - P(X \leq 3) = 1 - \sum_{k=0}^3 \frac{e^{-\lambda} \lambda^k}{k!} \\
 &= 1 - \left(\frac{e^{-2.5} (2.5)^0}{0!} + \frac{e^{-2.5} (2.5)^1}{1!} + \frac{e^{-2.5} (2.5)^2}{2!} + \frac{e^{-2.5} (2.5)^3}{3!} \right) \approx 0.1859
 \end{aligned}$$

Assignment 6

Prob Stats 1: Assignment Solutions

Eklavya Raman - 24MS024

1. We have an irregular six-sided die with faces numbered 1 to 6. The probability of 5 even numbers in 10 throws is twice that of getting 4 even numbers. If the die is thrown in 10^5 sets of 10 throws each, we need to find the expected number of sets with no even numbers. Let the probability of getting an even number be p and the probability of getting an odd number be q . We have -

$$P(5 \text{ even numbers}) = \binom{10}{5} p^5 q^5$$

$$P(4 \text{ even numbers}) = \binom{10}{4} p^4 q^6$$

We are given that -

$$\binom{10}{5} p^5 q^5 = 2 \cdot \binom{10}{4} p^4 q^6$$

Simplifying, we get -

$$\frac{10!}{5!5!} p^5 q^5 = 2 \cdot \frac{10!}{4!6!} p^4 q^6$$

$$\frac{1}{5!} p q^5 = 2 \cdot \frac{1}{4!6!} q^6$$

$$p = \frac{2 \cdot 5!}{4!6!} q$$

$$p = \frac{10}{6} q$$

Since $p + q = 1$, we have -

$$\frac{10}{6} q + q = 1$$

$$\frac{16}{6} q = 1$$

$$q = \frac{6}{16} = \frac{3}{8}$$

Thus, we have -

$$p = 1 - q = 1 - \frac{3}{8} = \frac{5}{8}$$

The probability of getting no even numbers in 10 throws is given by -

$$P(0 \text{ even numbers}) = \binom{10}{0} p^0 q^{10} = (1)(1) \left(\frac{3}{8}\right)^{10} = \left(\frac{3}{8}\right)^{10}$$

Therefore, the expected number of sets with no even numbers in 10^5 sets of 10 throws each is given by -

$$E(\text{sets with no even numbers}) = 10^5 \cdot P(0 \text{ even numbers}) = 10^5 \cdot \left(\frac{3}{8}\right)^{10}$$

$$= 100000 \cdot \left(\frac{3}{8}\right)^{10} \approx 5.5$$

2. We have the probability of missile hitting the target as 0.8. We need two strikes to destroy the target. We need to find the number of missiles such that the probability of destroying the target is at least 0.99. Let n be the number of missiles fired. The probability of destroying the target is given by -

$$\begin{aligned} P(\text{destroying target}) &= P(\text{at least 2 hits}) = 1 - P(0 \text{ hits}) - P(1 \text{ hit}) \\ &= 1 - \binom{n}{0}(0.8)^0(0.2)^n - \binom{n}{1}(0.8)^1(0.2)^{n-1} \\ &= 1 - (1)(1)(0.2)^n - (n)(0.8)(0.2)^{n-1} \end{aligned}$$

We need to find the smallest n such that -

$$\begin{aligned} 1 - (0.2)^n - 0.8n(0.2)^{n-1} &\geq 0.99 \\ (0.2)^n + 0.8n(0.2)^{n-1} &\leq 0.01 \end{aligned}$$

Testing for different values of n , we find that the smallest n satisfying the above inequality is 8. Thus, we have -

$$\boxed{n = 8}$$

3. We toss an unbiased coin until we get a head. We need to find the expected number of tosses required to get the first head. Let X be the random variable denoting the number of tosses required to get the first head. We have -

$$P(X = n) = \left(\frac{1}{2}\right)^n$$

The expected number of tosses required to get the first head is given by -

$$E(X) = \sum_{n=1}^{\infty} n \cdot P(X = n) = \sum_{n=1}^{\infty} n \cdot \left(\frac{1}{2}\right)^n$$

Since this is an infinite geometric series, we have -

$$E(X) = \frac{\frac{1}{2}}{(1 - \frac{1}{2})^2} = \frac{\frac{1}{2}}{(\frac{1}{2})^2} = \frac{\frac{1}{2}}{\frac{1}{4}} = 2$$

$$\boxed{E(X) = 2}$$

4. We have a box with a total of N balls and a white balls. We need to find the expected number of draws to get r white balls, drawing without replacement. We have $\binom{a}{r-1}$ ways to choose the first $r-1$ white balls from the a white balls and $\binom{N-a}{k-(r-1)}$ ways to choose the remaining $k-(r-1)$ balls from the $N-a$ non-white balls. Finally, we have $a-(r-1)$ ways to choose the r^{th} white ball from the remaining white balls. Thus, we have the pmf as -

$$\boxed{P(X = k) = \frac{\binom{a}{r-1} \cdot \binom{N-a}{k-(r-1)} \cdot (a-(r-1))}{\binom{N}{k}}}$$

5. The number of telephone calls arriving at an exchange between 9 AM and 10 AM, 10 AM and 11 AM, are Poisson random variables with parameters $\lambda = 2$ and $\lambda = 4$ respectively. The variables are independent. We need to find the probability that at least 3 calls arrive between 9 AM and 11 AM. Let X be the random variable denoting the number of calls arriving between 9 AM and 10 AM and Y be the random variable denoting the number of calls arriving between 10 AM and 11 AM. We have -

$$X \sim \text{Poisson}(\lambda = 2)$$

$$Y \sim \text{Poisson}(\lambda = 4)$$

We need to find $P(X + Y \geq 3)$. Since the sum of independent Poisson random variables is also a Poisson random variable with parameter equal to the sum of the individual parameters, we have -

$$X + Y \sim \text{Poisson}(\lambda = 2 + 4 = 6)$$

Thus, we have -

$$P(X + Y \geq 3) = 1 - P(X + Y < 3) = 1 - \sum_{k=0}^2 \frac{e^{-\lambda} \lambda^k}{k!}$$

$$\boxed{= 1 - \left(\frac{e^{-6}(6)^0}{0!} + \frac{e^{-6}(6)^1}{1!} + \frac{e^{-6}(6)^2}{2!} \right) \approx 0.9380}$$

6. We have a bimodal poisson distribution with modes at 1 and 2. We need to find first the value of λ and then the probability that $P(X = 1 \text{ or } X = 2)$. We know that the modes of a Poisson distribution occur at $\lfloor \lambda \rfloor$ and $\lfloor \lambda \rfloor - 1$ if λ is an integer. Since the modes are at 1 and 2, we have $\lambda = 2$. Thus, we have -

$$P(X = 1 \text{ or } X = 2) = P(X = 1) + P(X = 2)$$

$$= \frac{e^{-\lambda} \lambda^1}{1!} + \frac{e^{-\lambda} \lambda^2}{2!}$$

$$\boxed{= \frac{e^{-2}(2)^1}{1!} + \frac{e^{-2}(2)^2}{2!} = 2e^{-2} + 2e^{-2} = 4e^{-2} \approx 0.5413}$$

7. X, Y are two *iid* geometric random variables. We need to find the conditional distribution of $X \mid X + Y = n$. We have -

$$P(X = k \mid X + Y = n) = \frac{P(X = k, X + Y = n)}{P(X + Y = n)} = \frac{P(X = k, Y = n - k)}{P(X + Y = n)}$$

$$= \frac{P(X = k)P(Y = n - k)}{P(X + Y = n)}$$

$$= \frac{(1 - p)^{k-1}p \cdot (1 - p)^{n-k-1}p}{P(X + Y = n)}$$

$$= \frac{(1 - p)^{n-2}p^2}{P(X + Y = n)}$$

We need to find $P(X + Y = n)$. We have -

$$P(X + Y = n) = \sum_{k=1}^{n-1} P(X = k)P(Y = n - k)$$

$$= \sum_{k=1}^{n-1} (1 - p)^{k-1}p \cdot (1 - p)^{n-k-1}p$$

$$= \sum_{k=1}^{n-1} (1 - p)^{n-2}p^2 = (n - 1)(1 - p)^{n-2}p^2$$

Thus, we have -

$$P(X = k \mid X + Y = n) = \frac{(1 - p)^{n-2}p^2}{(n - 1)(1 - p)^{n-2}p^2} = \frac{1}{n - 1}$$

Therefore, the conditional distribution of $X \mid X + Y = n$ is given by -

$$\boxed{P(X = k \mid X + Y = n) = \frac{1}{n - 1}, \quad k = 1, 2, \dots, n - 1}$$

8. Let us assume that both match boxes have N matches each. We need to find the probability that if he selects a pocket at random he finds one pocket empty and the other containing r matches. Consider taking matches from both pockets until one pocket is empty. Define taking a match from the left pocket (L) as a success and taking a match from the right pocket (R) as a failure. Let us denote the event that one pocket (WLOG L) is empty and the other has r matches by E . We need to find the probability of getting N successes before getting $N - r$ failures. We have -

- The number of ways of arranging N successes and $N - r - 1$ failures is given by $\binom{2N-r-1}{N}$.
- The probability of each such arrangement is given by $p^N q^{N-r-1}$ where p is the probability of taking a match from the left pocket and q is the probability of taking a match from the right pocket.
- Thus, the total probability of getting N successes before getting $N - r$ failures is given by -

$$P(E) = \binom{2N-r-1}{N} p^N q^{N-r-1} + \binom{2N-r-1}{N} q^N p^{N-r-1}$$

$$= \binom{2N-r-1}{N} (p^N q^{N-r-1} + q^N p^{N-r-1})$$

9. A factory produce 100 lightbulbs, of which 10 are defective. 15 lightbulbs are selected at random without replacement. We need to find the probability that exactly 2 lightbulbs are defective. Let the events be defined as follows: D : Lightbulb is defective, N : Lightbulb is non-defective, X : Number of defective lightbulbs selected

We have -

- Total ways of selecting 15 lightbulbs from 100 lightbulbs = $\binom{100}{15}$
- Ways of selecting 2 defective lightbulbs from 10 defective lightbulbs = $\binom{10}{2}$
- Ways of selecting 13 non-defective lightbulbs from 90 non-defective lightbulbs = $\binom{90}{13}$

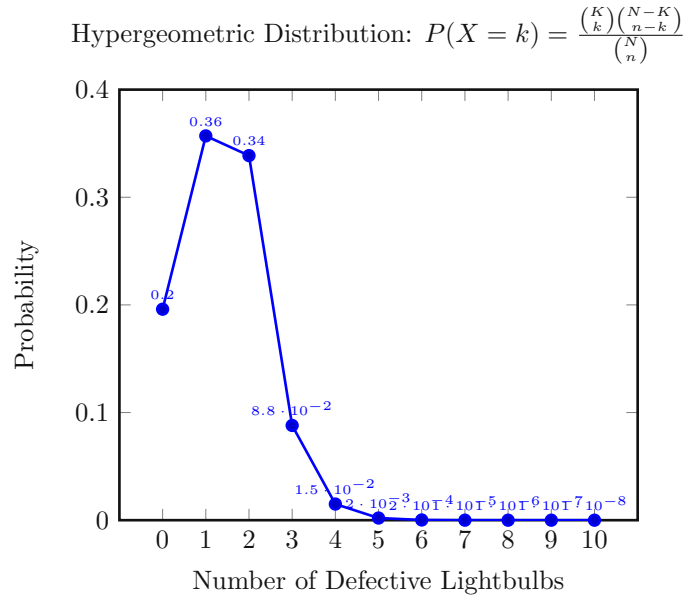
Therefore, the probability that exactly 2 lightbulbs are defective is given by -

$$P(X = 2) = \frac{\binom{10}{2} \cdot \binom{90}{13}}{\binom{100}{15}}$$

$$= \frac{45 \cdot 1905780240}{253338471349988640} \approx 0.3387$$

Note 6.0.0.1

Note: The distribution of X is hypergeometric with parameters $N = 100$, $K = 10$, $n = 15$. The plot of this distribution is shown below.



10. A fair die is thrown 10 times. We need to find the probability that each of 1,2,3,4 appear twice while 5 and 6 appear once each. Let the events be defined as follows: A_i : Face i appears on the die.

We have -

-
- Total number of outcomes when a fair die is thrown 10 times = 6^{10}
 - Number of favorable outcomes where each of 1,2,3,4 appear twice while 5 and 6 appear once each = $\frac{10!}{2!2!2!2!1!1!}$
 - Thus, the probability that each of 1,2,3,4 appear twice while 5 and 6 appear once each is given by -

$$P = \frac{10!}{2!2!2!2!1!1!6^{10}}$$

$= \frac{113400}{60466176} \approx 0.001874$
--

Assignment 7

Prob Stats 1: Assignment Solutions

Eklavya Raman - 24MS024

1. We have the frequency table as follows -

Expenditure in Rs.	40-59	60 -79	80-99	100-119	120-139
No. of families	50	x_1	500	x_2	50

(a) We have the mean given as Rs. 87.5. We need to find the values of x_1 and x_2 . We have -

$$\mu = \frac{\sum f_i x_i}{\sum f_i}$$

where f_i is the frequency and x_i is the mid-point of the class interval. Thus, we have -

$$87.5 = \frac{50 \cdot 49.5 + x_1 \cdot 69.5 + 500 \cdot 89.5 + x_2 \cdot 109.5 + 50 \cdot 129.5}{50 + x_1 + 500 + x_2 + 50}$$

$$87.5(600 + x_1 + x_2) = 2475 + 69.5x_1 + 44750 + 109.5x_2 + 6475$$

$$52500 + 87.5x_1 + 87.5x_2 = 53700 + 69.5x_1 + 109.5x_2$$

$$18x_1 - 22x_2 = 1200$$

$$9x_1 - 11x_2 = 600 \quad \dots (1)$$

We also have the total number of families as 1000. Thus, we have -

$$50 + x_1 + 500 + x_2 + 50 = 1000$$

$$x_1 + x_2 = 400 \quad \dots (2)$$

Solving equations (1) and (2), we get -

$$x_1 = 220, \quad x_2 = 180$$

$$(x_1, x_2) = (220, 180)$$

(b) We have the median given as Rs. 87.5. We need to find the values of x_1 and x_2 . We have the cumulative frequency table as follows -

Expenditure in Rs.	40-59	60 -79	80-99	100-119	120-139
No. of families	50	x_1	500	x_2	50
Cumulative frequency	50	$50 + x_1$	$550 + x_1$	$550 + x_1 + x_2$	1000

We know that the median class is the class where the cumulative frequency is greater than or equal to $\frac{N}{2}$. Here, $N = 1000$. Thus, we have -

$$\frac{N}{2} = 500$$

Thus, the median class is 80-99. We have -

$$\text{Median} = L + \left(\frac{\frac{N}{2} - F}{f_m} \right) \cdot h$$

where L is the lower limit of the median class, N is the total number of observations, F is the cumulative frequency of the class preceding the median class, f_m is the frequency of the median class and h is the class width. Thus, we have -

$$87.5 = 79.5 + \left(\frac{500 - (50 + x_1)}{500} \right) \cdot 20$$

$$8 = \left(\frac{450 - x_1}{500} \right) \cdot 20$$

$$2000 = 9000 - 20x_1$$

$$20x_1 = 7000$$

$$x_1 = 350 \quad \dots (1)$$

We also have the total number of families as 1000. Thus, we have -

$$50 + x_1 + 500 + x_2 + 50 = 1000$$

$$x_1 + x_2 = 400 \quad \dots (2)$$

Solving equations (1) and (2), we get -

$$x_1 = 350, \quad x_2 = 50$$

$$(x_1, x_2) = (350, 50)$$

- (c) We have the mode given as Rs. 87.5. We need to find the values of x_1 and x_2 . We have the cumulative frequency table as follows -

Expenditure in Rs.	40-59	60 -79	80-99	100-119	120-139
No. of families	50	x_1	500	x_2	50
Cumulative frequency	50	$50 + x_1$	$550 + x_1$	$550 + x_1 + x_2$	1000

We know that the modal class is the class with the highest frequency. Here, the modal class is 80-99. We have -

$$\text{Mode} = L + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \cdot h$$

where L is the lower limit of the modal class, f_1 is the frequency of the modal class, f_0 is the frequency of the class preceding the modal class, f_2 is the frequency of the class succeeding the modal class and h is the class width. Thus, we have -

$$87.5 = 79.5 + \left(\frac{500 - x_1}{2 \cdot 500 - x_1 - x_2} \right) \cdot 20$$

$$8 = \left(\frac{500 - x_1}{1000 - x_1 - x_2} \right) \cdot 20$$

$$4000 = 10000 - 20x_1 - 20x_2$$

$$20x_1 + 20x_2 = 6000$$

$$x_1 + x_2 = 300 \quad \dots (1)$$

We also have the total number of families as 1000. Thus, we have -

$$50 + x_1 + 500 + x_2 + 50 = 1000$$

$$x_1 + x_2 = 400 \quad \dots (2)$$

Solving equations (1) and (2), we get a contradiction. Thus, there is no solution for x_1 and x_2 such that the mode is Rs. 87.5.

$$\text{No solution for } (x_1, x_2)$$

2. We have that the values of a variable are in G.P, i.e., $a, ar, ar^2, ar^3, \dots, ar^{n-1}$. We need to show that A.M., G.M. and H.M. are in G.P. We have -

(a) Arithmetic Mean (A.M.) is given by -

$$\begin{aligned} A.M. &= \frac{a + ar + ar^2 + ar^3 + \dots + ar^{n-1}}{n} = \frac{a(1 + r + r^2 + r^3 + \dots + r^{n-1})}{n} \\ &= \frac{a \left(\frac{r^n - 1}{r - 1} \right)}{n} = \frac{a(r^n - 1)}{n(r - 1)} \end{aligned}$$

(b) Geometric Mean (G.M.) is given by -

$$G.M. = (a \cdot ar \cdot ar^2 \cdot ar^3 \cdot \dots \cdot ar^{n-1})^{\frac{1}{n}} = (a^n r^{\frac{n(n-1)}{2}})^{\frac{1}{n}} = ar^{\frac{n-1}{2}}$$

(c) Harmonic Mean (H.M.) is given by -

$$\begin{aligned} H.M. &= \frac{n}{\frac{1}{a} + \frac{1}{ar} + \frac{1}{ar^2} + \frac{1}{ar^3} + \dots + \frac{1}{ar^{n-1}}} = \frac{n}{\frac{1}{a} \left(1 + \frac{1}{r} + \frac{1}{r^2} + \dots + \frac{1}{r^{n-1}} \right)} \\ &= \frac{na}{\left(\frac{1 - \frac{1}{r^n}}{1 - \frac{1}{r}} \right)} = \frac{na(r - 1)r^{n-1}}{r^n - 1} \end{aligned}$$

We need to show that A.M., G.M. and H.M. are in G.P. Thus, we need to show that -

$$(G.M.)^2 = A.M. \cdot H.M.$$

We have -

$$\begin{aligned} (G.M.)^2 &= (ar^{\frac{n-1}{2}})^2 = a^2 r^{n-1} \\ A.M. \cdot H.M. &= \left(\frac{a(r^n - 1)}{n(r - 1)} \right) \cdot \left(\frac{na(r - 1)r^{n-1}}{r^n - 1} \right) = a^2 r^{n-1} \end{aligned}$$

Thus, we have shown that -

$$(G.M.)^2 = A.M. \cdot H.M.$$

Therefore, A.M., G.M. and H.M. are in G.P.

3. We have the frequency of a variable assuming value i for $i = 1, 2, \dots, 7$ is given by $f(i) = i^2$. We need to find the mean of the variable, that is $E(X)$. We have -

$$E(X) = \frac{\sum_{i=1}^7 i \cdot f(i)}{\sum_{i=1}^7 f(i)} = \frac{\sum_{i=1}^7 i \cdot i^2}{\sum_{i=1}^7 i^2} = \frac{\sum_{i=1}^7 i^3}{\sum_{i=1}^7 i^2}$$

We know that -

$$\begin{aligned} \sum_{i=1}^n i^2 &= \frac{n(n+1)(2n+1)}{6} \\ \sum_{i=1}^n i^3 &= \left(\frac{n(n+1)}{2} \right)^2 \end{aligned}$$

Thus, we have -

$$E(X) = \frac{\left(\frac{7 \cdot 8}{2} \right)^2}{\frac{7 \cdot 8 \cdot 15}{6}} = \frac{(28)^2}{140} = \frac{784}{140} = \frac{56}{10} = 5.6$$

$$\boxed{E(X) = 5.6}$$

4. We have the A.M. of $x_1, x_2, \dots, x_5$, with $x_i \geq x_j \iff i \geq j$ as 10. We have $y_1 = 14, y_2 = x_1, y_3 = y_4 = x_2, y_5 = y_6 = x_3, y_7 = y_8 = x_4, y_9 = y_{10} = x_5$. We need to show that the A.M. of $y_1, y_2, \dots, y_{10}$ cannot be less than 10.4. We have -

$$A.M. = \frac{x_1 + x_2 + x_3 + x_4 + x_5}{5} = 10$$

$$x_1 + x_2 + x_3 + x_4 + x_5 = 50$$

We need to find the A.M. of $y_1, y_2, \dots, y_{10}$. Thus, we have -

$$\begin{aligned} A.M. &= \frac{y_1 + y_2 + y_3 + y_4 + y_5 + y_6 + y_7 + y_8 + y_9 + y_{10}}{10} \\ &= \frac{14 + x_1 + x_2 + x_2 + x_3 + x_3 + x_4 + x_4 + x_5 + x_5}{10} \\ &= \frac{14 + x_1 + 2x_2 + 2x_3 + 2x_4 + 2x_5}{10} \\ &= \frac{14 + (x_1 + x_2 + x_3 + x_4 + x_5) + (x_2 + x_3 + x_4 + x_5)}{10} \\ &= \frac{14 + 50 + (x_2 + x_3 + x_4 + x_5)}{10} \\ &= \frac{64 + (x_2 + x_3 + x_4 + x_5)}{10} \end{aligned}$$

Since $x_1 \leq x_2 \leq x_3 \leq x_4 \leq x_5$, we have -

$$x_2 + x_3 + x_4 + x_5 \geq 4x_1$$

Thus, we have -

$$A.M. \geq \frac{64 + 4x_1}{10}$$

Since $x_1 + x_2 + x_3 + x_4 + x_5 = 50$, we have -

$$x_1 \leq 10$$

Thus, we have -

$$A.M. \geq \frac{64 + 4 \cdot 10}{10} = \frac{64 + 40}{10} = \frac{104}{10} = 10.4$$

$$\boxed{A.M. \geq 10.4}$$

5. The mean of 50 observations is 18. Values 22 and 16 were wrongly copied as 19 and 24 respectively. We need to find the correct mean. We have -

$$\text{Correct Sum} = \text{Original Sum} - \text{Wrong Values} + \text{Correct Values}$$

$$= 50 \cdot 18 - (19 + 24) + (22 + 16)$$

$$= 900 - 43 + 38 = 895$$

$$\text{Correct Mean} = \frac{\text{Correct Sum}}{50} = \frac{895}{50} = 17.9$$

$$\boxed{\text{Correct Mean} = 17.9}$$

6. If the average monthly income of factory workers is Rs. 5200 and average income of male workers is Rs. 5400 and Rs. 4600 respectively. We need to find the percentage of male and female workers in the factory. Let the number of male workers be m and the number of female workers be f . We have -

$$5200(m + f) = 5400m + 4600f$$

$$5200m + 5200f = 5400m + 4600f$$

$$5200f - 4600f = 5400m - 5200m$$

$$600f = 200m$$

$$\frac{m}{f} = \frac{600}{200} = 3$$

Thus, we have -

(a) Percentage of male workers -

$$\frac{m}{m+f} \cdot 100 = \frac{3f}{3f+f} \cdot 100 = \frac{3}{4} \cdot 100 = 75\%$$

$$\boxed{75\%}$$

(b) Percentage of female workers -

$$\frac{f}{m+f} \cdot 100 = \frac{f}{3f+f} \cdot 100 = \frac{1}{4} \cdot 100 = 25\%$$

$$\boxed{25\%}$$

7. A variable x takes the values $1, 2, 3, \dots, k$ with cumulative frequencies $F_1, F_2, F_3, \dots, F_k$ respectively. We need to show that the mean of the variable is given by -

$$\text{Mean} = \frac{1}{N} \sum F_i$$

where N is the total number of observations.

Proof. We know that the mean of the variable is given by -

$$\text{Mean} = \frac{1}{N} \sum_{i=1}^k x_i f_i$$

where f_i is the frequency of the variable taking value x_i . We have -

$$f_1 = F_1$$

$$f_2 = F_2 - F_1$$

$$f_3 = F_3 - F_2$$

$$\vdots$$

$$f_k = F_k - F_{k-1}$$

Thus, we have -

$$\begin{aligned} \text{Mean} &= \frac{1}{N} (1 \cdot F_1 + 2 \cdot (F_2 - F_1) + 3 \cdot (F_3 - F_2) + \dots + k \cdot (F_k - F_{k-1})) \\ &= \frac{1}{N} (F_1 + 2F_2 - 2F_1 + 3F_3 - 3F_2 + \dots + kF_k - kF_{k-1}) \\ &= \frac{1}{N} (kF_k + (2-1)F_2 + (3-2)F_3 + \dots + (k-(k-1))F_{k-1} - F_1) \\ &= \frac{1}{N} (kF_k + F_2 + F_3 + \dots + F_{k-1} - F_1) \\ &= \frac{1}{N} (F_1 + F_2 + F_3 + \dots + F_k) \end{aligned}$$

$$\boxed{= \frac{1}{N} \sum F_i}$$

□

8. We have a frequency table with intervals of 10 marks each. The median of the distribution is 35.5. On verification, two observations belonging to the interval 31-40 were included in the class 21-30. After correction, the median was found to be 36.5. We need to calculate the corrected frequency of the class 31-40. Let the frequency of the class 21-30 be f_1 and the frequency of the class 31-40 be f_2 . We have the median formula as -

$$\text{Median} = L + \left(\frac{\frac{N}{2} - F}{f} \right) \cdot h$$

where L is the lower class boundary of the median class, N is the total number of observations, F is the cumulative frequency of the class preceding the median class, f is the frequency of the median class and h is the class width.

For the initial median of 35.5, we have -

$$\begin{aligned} 35.5 &= 31 + \left(\frac{\frac{N}{2} - (F + f_1)}{f_2} \right) \cdot 10 \\ \Rightarrow 4.5 &= \left(\frac{\frac{N}{2} - F - f_1}{f_2} \right) \cdot 10 \\ \Rightarrow \frac{N}{2} - F - f_1 &= \frac{4.5f_2}{10} = 0.45f_2 \quad (1) \end{aligned}$$

For the corrected median of 36.5, we have -

$$\begin{aligned} 36.5 &= 31 + \left(\frac{\frac{N}{2} - (F + f_1 - 2)}{f_2 + 2} \right) \cdot 10 \\ \Rightarrow 5.5 &= \left(\frac{\frac{N}{2} - F - f_1 + 2}{f_2 + 2} \right) \cdot 10 \\ \Rightarrow \frac{N}{2} - F - f_1 + 2 &= \frac{5.5(f_2 + 2)}{10} = 0.55f_2 + 1.1 \quad (2) \end{aligned}$$

Subtracting (1) from (2), we get -

$$\begin{aligned} 2 &= (0.55f_2 + 1.1) - (0.45f_2) \\ 2 &= 0.1f_2 + 1.1 \\ \Rightarrow 0.9 &= 0.1f_2 \\ \Rightarrow f_2 &= 9 \end{aligned}$$

Therefore, the corrected frequency of the class 31-40 is given by -

$$\boxed{f_2 + 2 = 9 + 2 = 11}$$

9. The upper boundary of each class interval has a constant ratio to the lower boundary. Show that the G.M. may be expressed as

$$\ln G = A + \frac{k}{n} \sum_{i=1}^r (i-1)f_i$$

Proof. Let the lower boundary of the first class interval be a and the common ratio be r . Thus, we have the class intervals as -

- Class 1: a to ar
- Class 2: ar to ar^2
- Class 3: ar^2 to ar^3
- ...

- Class i : ar^{i-1} to ar^i

Let the frequency of class i be f_i . We have the G.M. as -

$$\ln G = \frac{1}{N} \sum_{i=1}^r f_i \ln x_i$$

where x_i is the mid-point of class i . We have -

$$x_i = \frac{ar^{i-1} + ar^i}{2} = \frac{ar^{i-1}(1+r)}{2}$$

Thus, we have -

$$\begin{aligned} \ln G &= \frac{1}{N} \sum_{i=1}^r f_i \ln \left(\frac{ar^{i-1}(1+r)}{2} \right) \\ &= \frac{1}{N} \sum_{i=1}^r f_i \left(\ln a + (i-1) \ln r + \ln \frac{1+r}{2} \right) \\ &= \frac{1}{N} \left(\sum_{i=1}^r f_i \ln a + \sum_{i=1}^r f_i (i-1) \ln r + \sum_{i=1}^r f_i \ln \frac{1+r}{2} \right) \\ &= \frac{1}{N} \left(N \ln a + \ln r \sum_{i=1}^r f_i (i-1) + N \ln \frac{1+r}{2} \right) \\ &= \ln a + \ln \frac{1+r}{2} + \frac{\ln r}{N} \sum_{i=1}^r f_i (i-1) \end{aligned}$$

Let us denote $\ln a + \ln \frac{1+r}{2}$ by A and $\ln r$ by k . Thus, we have -

$$\ln G = A + \frac{k}{N} \sum_{i=1}^r (i-1) f_i$$

$$\boxed{= A + \frac{k}{n} \sum_{i=1}^r (i-1) f_i}$$

□

10. In a batch of 15, 5 students failed the test. The marks of students who passed were 90, 60, 70, 80, 80, 90, 60, 50, 40, 70. We need to find median marks of all the 15 students. We have the marks of the 10 students who passed the test as follows -

Marks	Frequency
40	1
50	1
60	2
70	2
80	2
90	2

We need to find the median of all 15 students. Thus, we have -

$$\text{Median} = L + \left(\frac{\frac{N}{2} - F}{f} \right) \cdot h$$

where L is the lower class boundary of the median class, N is the total number of observations, F is the cumulative frequency of the class preceding the median class, f is the frequency of the

median class and h is the class width.

Here, we have -

$$N = 15$$
$$\frac{N}{2} = \frac{15}{2} = 7.5$$

The median class is 60-70. Thus, we have -

$$L = 60$$

$$F = 2$$

$$f = 4$$

$$h = 10$$

Thus, we have -

$$\begin{aligned}\text{Median} &= 60 + \left(\frac{7.5 - 2}{4} \right) \cdot 10 \\ &= 60 + \left(\frac{5.5}{4} \right) \cdot 10 = 60 + 13.75 = 73.75\end{aligned}$$

Median = 73.75

Assignment 8

Prob Stats 1: Assignment Solutions

Eklavya Raman - 24MS024

1. A variable only takes values x_1, x_2 with equal frequencies. We need to find the mean deviation and standard deviation in terms of x_1 and x_2 .

(a) Mean Deviation: We have the mean of the variable as -

$$\mu = \frac{x_1 + x_2}{2}$$

Thus, we have the mean deviation as -

$$\begin{aligned} MD &= \frac{|x_1 - \mu| + |x_2 - \mu|}{2} = \frac{\left|x_1 - \frac{x_1 + x_2}{2}\right| + \left|x_2 - \frac{x_1 + x_2}{2}\right|}{2} \\ &= \frac{\left|\frac{x_1 - x_2}{2}\right| + \left|\frac{x_2 - x_1}{2}\right|}{2} = \frac{\frac{|x_1 - x_2|}{2} + \frac{|x_1 - x_2|}{2}}{2} = \frac{|x_1 - x_2|}{2} \end{aligned}$$

$$MD = \frac{|x_1 - x_2|}{2}$$

(b) Standard Deviation: We have the standard deviation of the variable as -

$$\begin{aligned} \sigma &= \sqrt{\frac{(x_1 - \mu)^2 + (x_2 - \mu)^2}{2}} = \sqrt{\frac{\left(x_1 - \frac{x_1 + x_2}{2}\right)^2 + \left(x_2 - \frac{x_1 + x_2}{2}\right)^2}{2}} \\ &= \sqrt{\frac{\left(\frac{x_1 - x_2}{2}\right)^2 + \left(\frac{x_2 - x_1}{2}\right)^2}{2}} = \sqrt{\frac{2\left(\frac{(x_1 - x_2)^2}{4}\right)}{2}} = \sqrt{\frac{(x_1 - x_2)^2}{4}} = \frac{|x_1 - x_2|}{2} \end{aligned}$$

$$\sigma = \frac{|x_1 - x_2|}{2}$$

2. We have $3x + 4y = 5$ and the mean deviation of x about its mean is 8. We need to find the mean deviation of y about its mean. We have the equation $3x + 4y = 5$. Thus, we have -

$$y = \frac{5 - 3x}{4}$$

Let the mean of x be μ_x and the mean of y be μ_y . Thus, we have -

$$\mu_y = \frac{5 - 3\mu_x}{4}$$

We need to find the mean deviation of y about its mean, i.e., MD_y . We have -

$$\begin{aligned} MD_y &= E(|y - \mu_y|) = E\left(\left|\frac{5 - 3x}{4} - \frac{5 - 3\mu_x}{4}\right|\right) = E\left(\left|\frac{-3(x - \mu_x)}{4}\right|\right) \\ &= \frac{3}{4}E(|x - \mu_x|) = \frac{3}{4}MD_x \end{aligned}$$

Given that $MD_x = 8$, we have -

$$MD_y = \frac{3}{4} \cdot 8 = 6$$

$$\boxed{MD_y = 6}$$

3. We have the number of observations as 100, with mean and sd as 40.1 and 5 respectively. It was later found that 50 was copied instead of the correct value 40. We need to find the correct mean and sd.

(a) Correct Mean: We have -

$$\text{Correct Sum} = \text{Original Sum} - \text{Wrong Value} + \text{Correct Value}$$

$$= 100 \cdot 40.1 - 50 + 40$$

$$= 4010 - 50 + 40 = 4000$$

$$\text{Correct Mean} = \frac{\text{Correct Sum}}{100} = \frac{4000}{100} = 40$$

$$\boxed{\text{Correct Mean} = 40}$$

(b) Correct Standard Deviation: We have the formula for standard deviation as -

$$\sigma = \sqrt{\frac{\sum x_i^2}{N} - \mu^2}$$

where x_i are the observations, N is the number of observations and μ is the mean. Thus, we have -

$$5 = \sqrt{\frac{\sum x_i^2}{100} - (40.1)^2}$$

$$25 = \frac{\sum x_i^2}{100} - 1608.01$$

$$\sum x_i^2 = 162833.01$$

Now, we need to find the correct value of $\sum x_i^2$. We have -

$$\text{Correct } \sum x_i^2 = \text{Original } \sum x_i^2 - (\text{Wrong Value})^2 + (\text{Correct Value})^2$$

$$= 162833.01 - 50^2 + 40^2$$

$$= 162833.01 - 2500 + 1600 = 161933.01$$

Thus, we have the correct standard deviation as -

$$\sigma = \sqrt{\frac{161933.01}{100} - (40)^2} = \sqrt{1619.3301 - 1600} = \sqrt{19.3301} \approx 4.396$$

$$\boxed{\text{Correct Standard Deviation} \approx 4.396}$$

4. We have the class intervals as 10 marks each, total number of students as 40. We have the variance and mean as 90 and 40 respectively. Two observations belonging to the interval 21-30 were included in the class 31-40. We need to find the correct variance and mean. Let the frequency of the class 21-30 be f_1 and the frequency of the class 31-40 be f_2 . We have the mean formula as -

$$\mu = \frac{\sum f_i x_i}{N}$$

where f_i is the frequency of class i , x_i is the mid-point of class i and N is the total number of observations.

For the initial mean of 40, we have -

$$40 = \frac{\sum f_i x_i}{40}$$

$$\Rightarrow \sum f_i x_i = 1600 \quad (1)$$

For the corrected mean, we have -

$$\mu = \frac{\sum f_i x_i - 2 \cdot 25 + 2 \cdot 35}{40} = \frac{\sum f_i x_i + 20}{40}$$

$$\Rightarrow \mu = \frac{1600 + 20}{40} = \frac{1620}{40} = 40.5$$

$$\boxed{\text{Correct Mean} = 40.5}$$

We have the variance formula as -

$$\sigma^2 = \frac{\sum f_i x_i^2}{N} - \mu^2$$

where f_i is the frequency of class i , x_i is the mid-point of class i and N is the total number of observations.

For the initial variance of 90, we have -

$$90 = \frac{\sum f_i x_i^2}{40} - (40)^2$$

$$\Rightarrow \sum f_i x_i^2 = (90 + 1600) \cdot 40 = 67600 \quad (2)$$

For the corrected variance, we have -

$$\sigma^2 = \frac{\sum f_i x_i^2 - 2 \cdot 25^2 + 2 \cdot 35^2}{40} - (40.5)^2$$

$$\Rightarrow \sigma^2 = \frac{67600 - 1250 + 2450}{40} - 1640.25$$

$$\Rightarrow \sigma^2 = \frac{68800}{40} - 1640.25 = 1720 - 1640.25 = 79.75$$

$$\boxed{\text{Correct Variance} = 79.75}$$

5. The mean square deviation around 1 and -1 is 3 and 7 respectively. We need to find the sd of the observations. Let the mean of the observations be μ . We have the mean square deviation around 1 as -

$$MSD_1 = \frac{\sum (x_i - 1)^2}{N} = 3$$

$$\Rightarrow \sum (x_i - 1)^2 = 3N$$

$$\Rightarrow \sum (x_i^2 - 2x_i + 1) = 3N$$

$$\Rightarrow \sum x_i^2 - 2 \sum x_i + N = 3N$$

$$\Rightarrow \sum x_i^2 - 2N\mu + N = 3N$$

$$\Rightarrow \sum x_i^2 = 2N\mu + 2N \quad (1)$$

We have the mean square deviation around -1 as -

$$MSD_{-1} = \frac{\sum (x_i + 1)^2}{N} = 7$$

$$\Rightarrow \sum (x_i + 1)^2 = 7N$$

$$\Rightarrow \sum (x_i^2 + 2x_i + 1) = 7N$$

$$\Rightarrow \sum x_i^2 + 2 \sum x_i + N = 7N$$

$$\Rightarrow \sum x_i^2 + 2N\mu + N = 7N$$

$$\Rightarrow \sum x_i^2 = -2N\mu + 6N \quad (2)$$

Solving equations (1) and (2), we get -

$$2N\mu + 2N = -2N\mu + 6N$$

$$\Rightarrow 4N\mu = 4N$$

$$\Rightarrow \mu = 1$$

Substituting the value of μ in equation (1), we get -

$$\sum x_i^2 = 2N \cdot 1 + 2N = 4N$$

We have the formula for standard deviation as -

$$\begin{aligned} \sigma &= \sqrt{\frac{\sum x_i^2}{N} - \mu^2} \\ &= \sqrt{\frac{4N}{N} - (1)^2} = \sqrt{4 - 1} = \sqrt{3} \\ \boxed{\sigma = \sqrt{3}} \end{aligned}$$

6. We have the distribution as follows -

x	0	1	2	3	4	5	6	7	8
f	4	10	13	21	23	20	18	9	2

We need to find the mean deviation around median for this distribution. We have the total number of observations as -

$$N = 4 + 10 + 13 + 21 + 23 + 20 + 18 + 9 + 2 = 120$$

We need to find the median of the distribution. Thus, we have -

$$\frac{N}{2} = \frac{120}{2} = 60$$

The median class is 4. Thus, we have -

$$L = 4$$

$$F = 4 + 10 + 13 + 21 = 48$$

$$f = 23$$

$$h = 1$$

Thus, we have -

$$\text{Median} = 4 + \left(\frac{60 - 48}{23} \right) \cdot 1 = 4 + \frac{12}{23} = \frac{92 + 12}{23} = \frac{104}{23} \approx 4.52$$

Now, we need to find the mean deviation around median. Thus, we have -

$$\begin{aligned} MD &= \frac{\sum f_i |x_i - \text{Median}|}{N} \\ &= \frac{4 \left| 0 - \frac{104}{23} \right| + 10 \left| 1 - \frac{104}{23} \right| + 13 \left| 2 - \frac{104}{23} \right| + 21 \left| 3 - \frac{104}{23} \right| + 23 \left| 4 - \frac{104}{23} \right| + 20 \left| 5 - \frac{104}{23} \right|}{120} \\ &\quad + \frac{18 \left| 6 - \frac{104}{23} \right| + 9 \left| 7 - \frac{104}{23} \right| + 2 \left| 8 - \frac{104}{23} \right|}{120} \quad (8.0.0.1) \\ &= \frac{4 \cdot \frac{104}{23} + 10 \cdot \frac{81}{23} + 13 \cdot \frac{58}{23} + 21 \cdot \frac{35}{23} + 23 \cdot \frac{12}{23} + 20 \cdot \frac{19}{23} + 18 \cdot \frac{42}{23} + 9 \cdot \frac{65}{23} + 2 \cdot \frac{88}{23}}{120} \\ &= \frac{\frac{416 + 810 + 754 + 735 + 276 + 380 + 756 + 585 + 176}{23}}{120} = \frac{\frac{4888}{23}}{120} = \frac{4888}{2760} = \frac{1222}{690} \approx 1.77 \quad (8.0.0.2) \end{aligned}$$

7. In a batch of 10 children, IQ of a dull boy is 36 below the average IQ of the batch. Show that the sd of IQ for all children cannot be less than 10.8. If the sd is 11.4, we need to find the sd without the dull boy. We have the mean IQ of the batch as μ . Thus, we have the IQ of the dull boy as $\mu - 36$. We need to show that the sd of IQ for all children cannot be less than 10.8. We have the formula for standard deviation as -

$$\sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{N}}$$

where x_i are the observations, N is the number of observations and μ is the mean. Thus, we have -

$$\begin{aligned}\sigma^2 &= \frac{\sum (x_i - \mu)^2}{N} \\ \Rightarrow \sum (x_i - \mu)^2 &= N\sigma^2 \\ \Rightarrow \sum (x_i - \mu)^2 &= 10\sigma^2 \quad (1)\end{aligned}$$

We know that -

$$(x_i - \mu)^2 \geq 0$$

Thus, we have -

$$\begin{aligned}(\mu - 36 - \mu)^2 &\leq \sum (x_i - \mu)^2 \\ \Rightarrow 1296 &\leq 10\sigma^2 \\ \Rightarrow \sigma^2 &\geq 129.6 \\ \Rightarrow \sigma &\geq \sqrt{129.6} = 10.8\end{aligned}$$

Thus, we have shown that the sd of IQ for all children cannot be less than 10.8. Now, we need to find the sd without the dull boy. We have the formula for standard deviation as -

$$\sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{N}}$$

where x_i are the observations, N is the number of observations and μ is the mean. We have the mean after removing the dull boy as -

$$\mu' = \frac{10\mu - (\mu - 36)}{9} = \frac{9\mu + 36}{9} = \mu + 4$$

We have the sum of squares after removing the dull boy as -

$$\begin{aligned}\sum (x_i - \mu')^2 &= \sum (x_i - \mu + \mu - \mu')^2 \\ &= \sum (x_i - \mu - 4)^2 = \sum (x_i - \mu)^2 - 8 \sum (x_i - \mu) + 16N \\ &= \sum (x_i - \mu)^2 + 16N \\ &= 10\sigma^2 + 160 \quad (2)\end{aligned}$$

Thus, we have the sd without the dull boy as -

$$\sigma' = \sqrt{\frac{\sum (x_i - \mu')^2}{N - 1}} = \sqrt{\frac{10\sigma^2 + 160}{9}}$$

Given that $\sigma = 11.4$, we have -

$$\sigma' = \sqrt{\frac{10 \cdot (11.4)^2 + 160}{9}} = \sqrt{\frac{10 \cdot 129.96 + 160}{9}} = \sqrt{\frac{1299.6 + 160}{9}} = \sqrt{\frac{1459.6}{9}} \approx \sqrt{162.18} \approx 12.74$$

$$\boxed{\sigma' \approx 12.74}$$

8. We have the Distribution as follows -

Age (in years)	20-25	25-30	30-35	35-40	40-45	45-50
No. of persons	110	90	75	55	40	30

We need to find the mean deviation about median and sd for the above distribution. We have the total number of persons as -

$$N = 110 + 90 + 75 + 55 + 40 + 30 = 400$$

We need to find the median of the distribution. Thus, we have -

$$\frac{N}{2} = \frac{400}{2} = 200$$

The median class is 30-35. Thus, we have -

$$L = 30$$

$$F = 110 + 90 = 200$$

$$f = 75$$

$$h = 5$$

Thus, we have -

$$\text{Median} = 30 + \left(\frac{200 - 200}{75} \right) \cdot 5 = 30 + 0 = 30$$

Now, we need to find the mean deviation about median. Thus, we have -

$$\begin{aligned} MD &= \frac{\sum f_i |x_i - \text{Median}|}{N} \\ &= \frac{110|22.5 - 30| + 90|27.5 - 30| + 75|32.5 - 30| + 55|37.5 - 30| + 40|42.5 - 30| + 30|47.5 - 30|}{400} \\ &= \frac{110 \cdot 7.5 + 90 \cdot 2.5 + 75 \cdot 2.5 + 55 \cdot 7.5 + 40 \cdot 12.5 + 30 \cdot 17.5}{400} \\ &= \frac{825 + 225 + 187.5 + 412.5 + 500 + 525}{400} = \frac{2675}{400} = 6.6875 \end{aligned}$$

$$\boxed{MD \approx 6.69}$$

Now, we need to find the sd for the distribution. We have the formula for standard deviation as -

$$\sigma = \sqrt{\frac{\sum f_i x_i^2}{N} - \mu^2}$$

where f_i is the frequency of class i , x_i is the mid-point of class i and N is the total number of observations.

We have the mean as -

$$\begin{aligned} \mu &= \frac{\sum f_i x_i}{N} = \frac{110 \cdot 22.5 + 90 \cdot 27.5 + 75 \cdot 32.5 + 55 \cdot 37.5 + 40 \cdot 42.5 + 30 \cdot 47.5}{400} \\ &= \frac{2475 + 2475 + 2437.5 + 2062.5 + 1700 + 1425}{400} = \frac{15575}{400} = 38.9375 \end{aligned}$$

Now, we have -

$$\begin{aligned} \sum f_i x_i^2 &= 110 \cdot (22.5)^2 + 90 \cdot (27.5)^2 + 75 \cdot (32.5)^2 + 55 \cdot (37.5)^2 + 40 \cdot (42.5)^2 + 30 \cdot (47.5)^2 \\ &= 110 \cdot 506.25 + 90 \cdot 756.25 + 75 \cdot 1056.25 + 55 \cdot 1406.25 + 40 \cdot 1806.25 + 30 \cdot 2256.25 \\ &= 55687.5 + 68062.5 + 79218.75 + 77343.75 + 72250 + 67687.5 = 420250 \end{aligned}$$

Thus, we have the sd as -

$$\sigma = \sqrt{\frac{420250}{400} - (38.9375)^2} = \sqrt{1050.625 - 1516.015625} \approx 7.29$$

$$\boxed{\sigma \approx 7.29}$$

9. We have two groups of 15, 22 with variances 9 and 16 respectively. We have that their means differ by 8.8. We need to find the sd of the combined group. We have the formula for standard deviation of combined group as -

$$\sigma_c = \sqrt{\frac{(n_1 - 1)\sigma_1^2 + (n_2 - 1)\sigma_2^2 + \frac{n_1 n_2}{n_1 + n_2}(\mu_1 - \mu_2)^2}{n_1 + n_2 - 1}}$$

where n_1, n_2 are the number of observations in group 1 and group 2 respectively, σ_1, σ_2 are the standard deviations of group 1 and group 2 respectively and μ_1, μ_2 are the means of group 1 and group 2 respectively.

Thus, we have -

$$\begin{aligned}\sigma_c &= \sqrt{\frac{(15 - 1) \cdot 9 + (22 - 1) \cdot 16 + \frac{15 \cdot 22}{15 + 22} \cdot (8.8)^2}{15 + 22 - 1}} \\ &= \sqrt{\frac{14 \cdot 9 + 21 \cdot 16 + \frac{330}{37} \cdot 77.44}{36}} \\ &= \sqrt{\frac{126 + 336 + 691.2}{36}} = \sqrt{\frac{1153.2}{36}} \approx \sqrt{32.0333} \approx 5.66 \\ &\quad \boxed{\sigma_c \approx 5.66}\end{aligned}$$

10. We need to show that if a single observation x is added to a set of n observations with mean $\bar{x}$ and variance s^2 , then the new variance will be larger than the old if

$$|x - \bar{x}| > s\sqrt{\frac{n+1}{n}}$$

Proof. Let the new mean after adding the observation x be $\bar{x}'$ and the new variance be s'^2 . We have -

$$\bar{x}' = \frac{n\bar{x} + x}{n+1}$$

We have the formula for variance as -

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n}$$

where x_i are the observations and n is the number of observations.

Thus, we have -

$$\begin{aligned}s'^2 &= \frac{\sum (x_i - \bar{x}')^2 + (x - \bar{x}')^2}{n+1} \\ &= \frac{\sum (x_i - \bar{x} + \bar{x} - \bar{x}')^2 + (x - \bar{x}')^2}{n+1} \\ &= \frac{\sum (x_i - \bar{x})^2 + 2(\bar{x} - \bar{x}') \sum (x_i - \bar{x}) + n(\bar{x} - \bar{x}')^2 + (x - \bar{x}')^2}{n+1} \\ &= \frac{ns^2 + n(\bar{x} - \bar{x}')^2 + (x - \bar{x}')^2}{n+1}\end{aligned}$$

We need to show that $s'^2 > s^2$. Thus, we have -

$$\begin{aligned}\frac{ns^2 + n(\bar{x} - \bar{x}')^2 + (x - \bar{x}')^2}{n+1} &> s^2 \\ \Rightarrow ns^2 + n(\bar{x} - \bar{x}')^2 + (x - \bar{x}')^2 &> (n+1)s^2 \\ \Rightarrow n(\bar{x} - \bar{x}')^2 + (x - \bar{x}')^2 &> s^2\end{aligned}$$

Substituting the value of $\bar{x}'$, we get -

$$n\left(\bar{x} - \frac{n\bar{x} + x}{n+1}\right)^2 + \left(x - \frac{n\bar{x} + x}{n+1}\right)^2 > s^2$$

$$\begin{aligned}
&\Rightarrow n \left(\frac{\bar{x} - x}{n+1} \right)^2 + \left(\frac{x - \bar{x}}{n+1} \right)^2 > s^2 \\
&\Rightarrow \frac{n(x - \bar{x})^2 + (x - \bar{x})^2}{(n+1)^2} > s^2 \\
&\Rightarrow \frac{(n+1)(x - \bar{x})^2}{(n+1)^2} > s^2 \\
&\Rightarrow \frac{(x - \bar{x})^2}{n+1} > s^2 \\
&\Rightarrow |x - \bar{x}| > s\sqrt{n+1} \\
&\Rightarrow |x - \bar{x}| > s\sqrt{\frac{n+1}{n}}
\end{aligned}$$

Thus, we have shown that if a single observation x is added to a set of n observations with mean $\bar{x}$ and variance s^2 , then the new variance will be larger than the old if

$$|x - \bar{x}| > s\sqrt{\frac{n+1}{n}}$$

□

Assignment 9
Prob Stats 1: Assignment Solutions
Eklavya Raman - 24MS024

1. The mean, mode and coefficient of variation of the weekly wages of a group of workers are Rs. 50, Rs. 58 and 40% respectively. We need to find the sd and median and coefficient of skewness of the weekly wages.

(a) Standard Deviation: We have the formula for coefficient of variation as -

$$CV = \frac{\sigma}{\mu} \cdot 100$$

where σ is the standard deviation and μ is the mean.
Substituting the values, we get -

$$40 = \frac{\sigma}{50} \cdot 100$$
$$\Rightarrow \sigma = \frac{40 \cdot 50}{100} = 20$$

Thus, the standard deviation is Rs. 20.

$$\boxed{\sigma = 20}$$

(b) Median: We have the formula for median in terms of mean and mode as -

$$\text{Median} = 3 \cdot \text{Mode} - 2 \cdot \text{Mean}$$
$$= 3 \cdot 58 - 2 \cdot 50 = 174 - 100 = 74$$

Thus, the median is Rs. 74.

$$\boxed{\text{Median} = 74}$$

(c) Coefficient of Skewness: We have the formula for coefficient of skewness as -

$$Sk = \frac{3(\mu - \text{Median})}{\sigma}$$
$$= \frac{3(50 - 74)}{20} = \frac{3(-24)}{20} = \frac{-72}{20} = -3.6$$

Thus, the coefficient of skewness is -3.6.

$$\boxed{Sk = -3.6}$$

2. In a moderately skewed distribution, the mean is 50, median is 53 and coefficient of variation is 20%. We need to find the value of coefficient of skewness. We have the formula for coefficient of skewness as -

$$Sk = \frac{3(\mu - \text{Median})}{\sigma}$$

We have the mean as $\mu = 50$ and median as 53. We need to find the standard deviation σ . We have the formula for coefficient of variation as -

$$CV = \frac{\sigma}{\mu} \cdot 100$$

Substituting the values, we get -

$$20 = \frac{\sigma}{50} \cdot 100$$

$$\Rightarrow \sigma = \frac{20 \cdot 50}{100} = 10$$

Now, substituting the values in the formula for coefficient of skewness, we get -

$$Sk = \frac{3(50 - 53)}{10} = \frac{3(-3)}{10} = \frac{-9}{10} = -0.9$$

Thus, the coefficient of skewness is -0.9.

$$\boxed{Sk = -0.9}$$

3. We have sd of a distribution as 3. We need to calculate the value of μ_4 such that the distribution is -

(a) Leptokurtic: We have the condition for leptokurtic distribution as -

$$\frac{\mu_4}{\sigma^4} > 3$$

where μ_4 is the fourth central moment and σ is the standard deviation.

Substituting the value of σ , we get -

$$\frac{\mu_4}{3^4} > 3$$

$$\Rightarrow \frac{\mu_4}{81} > 3$$

$$\Rightarrow \mu_4 > 243$$

Thus, for the distribution to be leptokurtic, μ_4 should be greater than 243.

$$\boxed{\mu_4 > 243}$$

(b) Mesokurtic: We have the condition for mesokurtic distribution as -

$$\frac{\mu_4}{\sigma^4} = 3$$

where μ_4 is the fourth central moment and σ is the standard deviation.

Substituting the value of σ , we get -

$$\frac{\mu_4}{3^4} = 3$$

$$\Rightarrow \frac{\mu_4}{81} = 3$$

$$\Rightarrow \mu_4 = 243$$

Thus, for the distribution to be mesokurtic, μ_4 should be equal to 243.

$$\boxed{\mu_4 = 243}$$

(c) Platykurtic: Similarly, we have the condition for platykurtic distribution as -

$$\frac{\mu_4}{\sigma^4} < 3$$

where μ_4 is the fourth central moment and σ is the standard deviation.

Same as parts (a) and (b), we get -

$$\mu_4 < 243$$

Thus, for the distribution to be platykurtic, μ_4 should be less than 243.

$$\boxed{\mu_4 < 243}$$

4. We need to verify whether the data is consistent or not. We have the data as follows -

- (a) $m_2 = 16, m_3 = -64, m_4 = 162$

We have the formula for central moments in terms of raw moments as -

$$\mu_2 = m_2 - m_1^2$$

$$\mu_3 = m_3 - 3m_1m_2 + 2m_1^3$$

$$\mu_4 = m_4 - 4m_1m_3 + 6m_1^2m_2 - 3m_1^4$$

where μ_n is the n^{th} central moment and m_n is the n^{th} raw moment.

We need to find the value of m_1 . We have the condition for consistency as -

$$\mu_4 \geq \mu_2^2$$

Substituting the values, we get -

$$\begin{aligned} 162 &\geq (16 - m_1^2)^2 \\ \Rightarrow 162 &\geq 256 - 32m_1^2 + m_1^4 \\ \Rightarrow m_1^4 - 32m_1^2 + 94 &\leq 0 \\ \Rightarrow (m_1^2 - 2)(m_1^2 - 30) &\leq 0 \\ \Rightarrow 2 &\leq m_1^2 \leq 30 \\ \Rightarrow \sqrt{2} &\leq |m_1| \leq \sqrt{30} \end{aligned}$$

Thus, the data is consistent for $|m_1|$ in the range

$$\sqrt{2}, \sqrt{30}$$

- (b) Mean = 63, Median = 65, coefficient of skewness = -0.375, Variance=576.

We have the formula for coefficient of skewness as -

$$Sk = \frac{3(\mu - \text{Median})}{\sigma}$$

where μ is the mean and σ is the standard deviation.

We have the standard deviation as -

$$\sigma = \sqrt{576} = 24$$

Substituting the values, we get -

$$\begin{aligned} -0.375 &= \frac{3(63 - 65)}{24} \\ -0.375 &= \frac{3(-2)}{24} \\ -0.375 &= -0.25 \end{aligned}$$

Thus, the data is not consistent.

Data is not consistent

- (c) $b_1 = 0.7, b_2 = 15$

We have the formula for coefficient of skewness in terms of moments as -

$$b_1 = \frac{\mu_3^2}{\mu_2^3}$$

$$b_2 = \frac{\mu_4}{\mu_2^2}$$

where μ_n is the n^{th} central moment.

We need to find the value of μ_2 . We have the condition for consistency as -

$$b_2 \geq b_1 + 1$$

Substituting the values, we get -

$$15 \geq 0.7 + 1$$

$$15 \geq 1.7$$

Thus, the data is consistent.

Data is consistent

5. The measure of skewness is 0.32. It is also known that the mean and variance are 29.6 and 42.25. We need to find the median and mode of the distribution. We have the formula for coefficient of skewness as -

$$Sk = \frac{3(\mu - \text{Median})}{\sigma}$$

where μ is the mean and σ is the standard deviation.

We have the standard deviation as -

$$\sigma = \sqrt{42.25} = 6.5$$

Substituting the values, we get -

$$0.32 = \frac{3(29.6 - \text{Median})}{6.5}$$

$$\Rightarrow 0.32 \cdot 6.5 = 3(29.6 - \text{Median})$$

$$\Rightarrow 2.08 = 88.8 - 3 \cdot \text{Median}$$

$$\Rightarrow 3 \cdot \text{Median} = 88.8 - 2.08 = 86.72$$

$$\Rightarrow \text{Median} = \frac{86.72}{3} \approx 28.91$$

Median ≈ 28.91

We have the formula for mode in terms of mean and median as -

$$\text{Mode} = 3 \cdot \text{Median} - 2 \cdot \mu$$

$$= 3 \cdot 28.91 - 2 \cdot 29.6$$

$$= 86.73 - 59.2 = 27.53$$

Mode ≈ 27.53

6. We have the mean as 120, mode as 123, and coefficient of skewness as -0.3. We need to find the coefficient of variation. We have the formula for coefficient of skewness as -

$$Sk = \frac{3(\mu - \text{Median})}{\sigma}$$

where μ is the mean and σ is the standard deviation.

We need to find the median first. Substituting the values, we get -

$$-0.3 = \frac{3(120 - \text{Median})}{\sigma}$$

$$\Rightarrow -0.3\sigma = 3(120 - \text{Median})$$

$$\Rightarrow -0.1\sigma = 120 - \text{Median}$$

$$\Rightarrow \text{Median} = 120 + 0.1\sigma \quad (1)$$

We have the formula for mode in terms of mean and median as -

$$\text{Mode} = 3 \cdot \text{Median} - 2 \cdot \mu$$

Substituting the values, we get -

$$123 = 3 \cdot \text{Median} - 2 \cdot 120$$

$$\Rightarrow 123 = 3 \cdot \text{Median} - 240$$

$$\Rightarrow 3 \cdot \text{Median} = 363$$

$$\Rightarrow \text{Median} = 121 \quad (2)$$

Solving equations (1) and (2), we get -

$$121 = 120 + 0.1\sigma$$

$$\Rightarrow 0.1\sigma = 1$$

$$\Rightarrow \sigma = 10$$

We have the formula for coefficient of variation as -

$$CV = \frac{\sigma}{\mu} \times 100$$

Substituting the values, we get -

$$CV = \frac{10}{120} \times 100 = \frac{1000}{120} \approx 8.33\%$$

$$\boxed{CV \approx 8.33\%}$$

7. We have that a variable X takes on only 2 values with equal frequency. We need to find μ_2, μ_3 and μ_4 in terms of a and b where the two values are a and b . Let the two values be a and b . We have the mean as -

$$\mu = \frac{a+b}{2}$$

We have the formula for n^{th} central moment as -

$$\mu_n = \frac{(a-\mu)^n + (b-\mu)^n}{2}$$

We now calculate μ_2, μ_3 and μ_4 as follows -

(a) μ_2 :

$$\begin{aligned} \mu_2 &= \frac{(a-\mu)^2 + (b-\mu)^2}{2} \\ &= \frac{\left(a - \frac{a+b}{2}\right)^2 + \left(b - \frac{a+b}{2}\right)^2}{2} \\ &= \frac{\left(\frac{a-b}{2}\right)^2 + \left(\frac{b-a}{2}\right)^2}{2} = \frac{\frac{(a-b)^2}{4} + \frac{(b-a)^2}{4}}{2} = \frac{\frac{(a-b)^2}{4} + \frac{(a-b)^2}{4}}{2} = \frac{\frac{2(a-b)^2}{4}}{2} = \frac{(a-b)^2}{4} \end{aligned}$$

$$\boxed{\mu_2 = \frac{(a-b)^2}{4}}$$

(b) μ_3 :

$$\begin{aligned} \mu_3 &= \frac{(a-\mu)^3 + (b-\mu)^3}{2} \\ &= \frac{\left(a - \frac{a+b}{2}\right)^3 + \left(b - \frac{a+b}{2}\right)^3}{2} \\ &= \frac{\left(\frac{a-b}{2}\right)^3 + \left(\frac{b-a}{2}\right)^3}{2} = \frac{\frac{(a-b)^3}{8} + \frac{(b-a)^3}{8}}{2} = \frac{\frac{(a-b)^3}{8} + \frac{-(a-b)^3}{8}}{2} = 0 \end{aligned}$$

$$\boxed{\mu_3 = 0}$$

(c) μ_4 :

$$\begin{aligned}
 \mu_4 &= \frac{(a - \mu)^4 + (b - \mu)^4}{2} \\
 &= \frac{\left(a - \frac{a+b}{2}\right)^4 + \left(b - \frac{a+b}{2}\right)^4}{2} \\
 &= \frac{\left(\frac{a-b}{2}\right)^4 + \left(\frac{b-a}{2}\right)^4}{2} = \frac{\frac{(a-b)^4}{16} + \frac{(b-a)^4}{16}}{2} = \frac{\frac{(a-b)^4}{16} + \frac{(a-b)^4}{16}}{2} = \frac{\frac{2(a-b)^4}{16}}{2} = \frac{(a-b)^4}{16}
 \end{aligned}$$

$$\mu_4 = \frac{(a-b)^4}{16}$$

8. We have the mean, variance and μ_3 of a distribution as 10, 16, 64. We need to find the values of the first four moments around the origin, given $b_2 = 4$. We have -

(a) m_1 : We have the first moment around the origin as the mean. Thus, we have -

$$m_1 = 10$$

(b) m_2 : We have the formula for variance in terms of moments as -

$$\sigma^2 = m_2 - m_1^2$$

where σ^2 is the variance, m_2 is the second moment around the origin and m_1 is the first moment around the origin.

Substituting the values, we get -

$$\begin{aligned}
 16 &= m_2 - 10^2 \\
 \Rightarrow m_2 &= 16 + 100 = 116
 \end{aligned}$$

$$m_2 = 116$$

(c) m_3 : We have the formula for third central moment in terms of moments as -

$$\mu_3 = m_3 - 3m_1m_2 + 2m_1^3$$

where μ_3 is the third central moment, m_3 is the third moment around the origin and m_1, m_2 are the first and second moments around the origin respectively.

Substituting the values, we get -

$$\begin{aligned}
 64 &= m_3 - 3 \cdot 10 \cdot 116 + 2 \cdot 10^3 \\
 \Rightarrow 64 &= m_3 - 3480 + 2000 \\
 \Rightarrow m_3 &= 64 + 3480 - 2000 = 1544
 \end{aligned}$$

$$m_3 = 1544$$

(d) m_4 : We have the formula for coefficient of kurtosis in terms of moments as -

$$b_2 = \frac{m_4 - 4m_1m_3 + 6m_1^2m_2 - 3m_1^4}{(m_2 - m_1^2)^2}$$

where b_2 is the coefficient of kurtosis, m_4 is the fourth moment around the origin and m_1, m_2, m_3 are the first, second and third moments around the origin respectively.

Substituting the values, we get -

$$\begin{aligned}
 4 &= \frac{m_4 - 4 \cdot 10 \cdot 1544 + 6 \cdot 10^2 \cdot 116 - 3 \cdot 10^4}{(116 - 10^2)^2} \\
 &= \frac{m_4 - 61760 + 69600 - 30000}{16^2} \\
 &= \frac{m_4 - 22160}{256} \\
 \Rightarrow 4 \cdot 256 &= m_4 - 22160 \\
 \Rightarrow m_4 &= 1024 + 22160 = 23184
 \end{aligned}$$

$$m_4 = 23184$$

9. We have the frequency table as follows -

Value	0	1	2	3	4	5	6	7	8
Frequency	1	8	28	56	70	56	28	8	1

We need to find the first four moments about the mean and hence find b_1, b_2 . We have the total frequency as -

$$N = 1 + 8 + 28 + 56 + 70 + 56 + 28 + 8 + 1 = 256$$

We have the mean as -

$$\begin{aligned}\mu &= \frac{\sum f_i x_i}{N} = \frac{1 \cdot 0 + 8 \cdot 1 + 28 \cdot 2 + 56 \cdot 3 + 70 \cdot 4 + 56 \cdot 5 + 28 \cdot 6 + 8 \cdot 7 + 1 \cdot 8}{256} \\ &= \frac{0 + 8 + 56 + 168 + 280 + 280 + 168 + 56 + 8}{256} = \frac{1024}{256} = 4\end{aligned}$$

Now, we calculate the first four moments about the mean as follows -

(a) μ_2 :

$$\begin{aligned}\mu_2 &= \frac{\sum f_i (x_i - \mu)^2}{N} \\ &= \frac{1 \cdot 16 + 8 \cdot 9 + 28 \cdot 4 + 56 \cdot 1 + 70 \cdot 0 + 56 \cdot 1 + 28 \cdot 4 + 8 \cdot 9 + 1 \cdot 16}{256} \\ &= \frac{16 + 72 + 112 + 56 + 0 + 56 + 112 + 72 + 16}{256} = \frac{512}{256} = 2 \\ &\quad \boxed{\mu_2 = 2}\end{aligned}$$

(b) μ_3 :

$$\begin{aligned}\mu_3 &= \frac{\sum f_i (x_i - \mu)^3}{N} \\ &= \frac{1 \cdot (-64) + 8 \cdot (-27) + 28 \cdot (-8) + 56 \cdot (-1) + 70 \cdot 0 + 56 \cdot 1 + 28 \cdot 8 + 8 \cdot 27 + 1 \cdot 64}{256} \\ &= \frac{-64 - 216 - 224 - 56 + 0 + 56 + 224 + 216 + 64}{256} = \frac{0}{256} = 0 \\ &\quad \boxed{\mu_3 = 0}\end{aligned}$$

(c) μ_4 :

$$\begin{aligned}\mu_4 &= \frac{\sum f_i (x_i - \mu)^4}{N} \\ &= \frac{1 \cdot 256 + 8 \cdot 81 + 28 \cdot 16 + 56 \cdot 1 + 70 \cdot 0 + 56 \cdot 1 + 28 \cdot 16 + 8 \cdot 81 + 1 \cdot 256}{256} \\ &= \frac{256 + 648 + 448 + 56 + 0 + 56 + 448 + 648 + 256}{256} = \frac{2816}{256} = 11 \\ &\quad \boxed{\mu_4 = 11}\end{aligned}$$

(d) b_1, b_2 : We have the formula for coefficient of skewness as -

$$\begin{aligned}b_1 &= \frac{\mu_3^2}{\mu_2^3} = \frac{0^2}{2^3} = 0 \\ &\quad \boxed{b_1 = 0}\end{aligned}$$

We have the formula for coefficient of kurtosis as -

$$\begin{aligned}b_2 &= \frac{\mu_4}{\mu_2^2} = \frac{11}{2^2} = \frac{11}{4} = 2.75 \\ &\quad \boxed{b_2 = 2.75}\end{aligned}$$

10. We have the following frequency distribution -

Class Interval	21-24	25-28	29-32	33-36	37-40	41-44
Frequency	110	90	75	55	40	30

We need to find the first four moments about the mean and hence find b_1, b_2 . We have the total frequency as -

$$N = 110 + 90 + 75 + 55 + 40 + 30 = 400$$

We have the midpoints of the class intervals as follows -

Class Interval	21-24	25-28	29-32	33-36	37-40	41-44
Midpoint	22.5	26.5	30.5	34.5	38.5	42.5

We have the mean as -

$$\begin{aligned}\mu &= \frac{\sum f_i x_i}{N} = \frac{110 \cdot 22.5 + 90 \cdot 26.5 + 75 \cdot 30.5 + 55 \cdot 34.5 + 40 \cdot 38.5 + 30 \cdot 42.5}{400} \\ &= \frac{2475 + 2385 + 2287.5 + 1897.5 + 1540 + 1275}{400} = \frac{13860}{400} = 34.65\end{aligned}$$

Now, we calculate the first four moments about the mean as follows -

(a) μ_2 :

$$\mu_2 = \frac{\sum f_i (x_i - \mu)^2}{N}$$

$\mu_2 = 64.505$

(b) μ_3 :

$$\mu_3 = \frac{\sum f_i (x_i - \mu)^3}{N}$$

$\mu_3 \approx -646.43$

(c) μ_4 :

$$\mu_4 = \frac{\sum f_i (x_i - \mu)^4}{N}$$

$\mu_4 \approx 7437.26$

We now have -

(a) b_1 : We have the formula for coefficient of skewness as -

$$b_1 = \frac{\mu_3^2}{\mu_2^3} \approx \frac{(-646.43)^2}{(64.505)^3} \approx \frac{417860.5}{268601.5} \approx 1.555$$

$b_1 \approx 1.555$

(b) b_2 : We have the formula for coefficient of kurtosis as -

$$b_2 = \frac{\mu_4}{\mu_2^2} \approx \frac{7437.26}{(64.505)^2} \approx \frac{7437.26}{4160.9} \approx 1.787$$

$b_2 \approx 1.787$